PALESTINIAN PHYSIOTHERAPY ASSOCIATION

LICENSING EXAMINATION

2010 -2011

1. A complete medical history should be conducted on all patients prior to initiating treatment . Questions asked during the patient history should not lead the patient . Which of the following questions would not be considered leading?

A. Does this increase your pain?

B. Does this alter your pain in any way ?

C. Does your pain increase at night ?

D. Does your pain increase with activity ?

2. A young boy is an insulin dependent diabetic . What effect does exercise have .on the patient's insulin requirements ?

A. Exercise often reduces a patient's insulin requirements .

B. Exercise often increases a patient's insulin requirements .

C. Exercise has no effect on a patient's insulin requirements .

D. Exercise is contraindicated for insulin dependent diabetics .

3. A patient with chronic obstructive pulmonary disease is referred to physical therapy .

Which of the following pulmonary function test results would you not expect to see when compared to normal values ?

A. Decreased lung volume .

B. Decreased aspiratory capacity .

C. **Increased vital capacity** .

D. Carbon dioxide retention in arterial blood .

4. A patient in a work hardening program is required to lift packages weighing approximately 30 pounds overhead to a conveyor belt .

The patient can complete the task, but is unable of the following assumptions is most accurate ?

A. Additional weight should be added to the packages which would promote lumbar stability .

B. The patient should continue lifting the 30 pound package because he will gradually become stronger .

C. The task is too easy for the patient .

D. The task is too difficult for the patient .

5. A patient recovering from a serious hamstring strain is isokinetically evaluated prior to returning to track competition . Results of the evaluation reveal peak torque measurements of 125 ft lbs with knee extension and 78 ft lbs with knee flexion at 180 degrees per second on the involved side . What conclusion can be made regarding the patient's ability to return to athletic competition?

A. The patient's quadriceps and hamstring strength are appropriate for a return to athletic activities .

B. The patient's hamstring strength demonstrates the need for continued rehabilitation .

C. The patient's quadriceps/hamstring ratio is below acceptable levels for athletic activities .

D. Not enough information is given to make accurate determination of the patient's functional status .

6. During an initial evaluation a physical therapist is looking for information regarding patient's general willingness to use an affected body part . What objective information would provide you with the necessary information?

A. Bony palpation .
B. Active movement .
C. Passive movement .
D. Sensory testing .

7. A therapist is treating a patient three weeks status post anterior cruciate ligament reconstruction. The patient has just seen their physician and has been told that he can discontinue the use of the straight leg immobilizer . The primary concern of the therapist at this stage in the patient's rehabilitation is ___?

A. Protection of the graft ?
B. Improvement in dynamic leg control during ambulation .
C. Increase quadriceps strength .
D. Improvement in knee range of motion particularly into extension

8. The ability to monitor vital signs is a necessary component of a physical therapy assessment . The purpose of obtaining vital signs include all of the following except ____?

A. Assisting in goal setting and treatment planning .
B. Contributing to the assessment of the effectiveness of treatment .
C. Determining the patient's rehabilitation potential.
D. Establishing a database of values for an individual patient .

9. A male physical therapist is evaluation a female diagnosed with subacromial bursitis . After taking a thorough history the therapist asks the patient to change into a gown so that he may begin to examine the shoulder . The patient seems very uneasy about this suggestion , but finally agrees to use the gown . The most appropriate immediate course of action would be to ___?

A. Continue the treatment without further delay .

B. Attempt to treat the patient without using the gown .

C. Bring a female staff member into the treatment room and continue with treatment .

D. Offer to transfer the patient to a female therapist .

10. improperly positioned wheelchair leg and footrests could result in ____?

A. Excessive pressure under the distal thigh .

B. Excessive pressure under the ischial tuberosities .

C. Contractures of the lower extremities .

D. All of the above .

11. A physical therapist is evaluating a patient diagnosed with an acute posterior cruciate sprain . The mechanism of injury for the posterior cruciate is __?

A. A forceful landing on the anterior tibia with the knee hyperflexed .

B. An anteriority directed force applied to the tibia when the foot is fixed .

C. A valgus force applied to the knee when the foot is fixed.

D. Hyperextension, internal rotation of the leg with external rotation of the body.

12. The knee bolt on an above knee prosthesis is usually rotated five degrees to account for ___?

A. **Internal rotation during swing phase** .

B. External rotation during swing phase .

C. Internal rotation during stance phase .

D. External rotation during stance phase .

13. practical and helpful recommendations to avoid unnecessary litigation include all of the following . perhaps the most important of these is ___?

A. To conduct a thorough initial evaluation .

B. To instruct patients carefully in all exercise activities .

C. To keep the referring physician informed .

D. **To maintain accurate and timely written records** .

14. A patient diagnosed with right shoulder adhesive capsulitis is limited to 25 degrees of external rotation . Which mobilization techniques would be indicated with this limitation ?

A. Lateral distraction and anterior glide .

B. Medial distraction and posterior glide .

C. Lateral distraction and posterior glide .

D. Medial distraction and inferior glide .

15. A former patient calls to ask for advice immediately after injuring his lower back in a work related accident . The patient explains that he can not bend down and touch his toes without severe pain and has muscle spasms throughout his entire lower back . You work in a state without direct access , but would like to help your former patient . The most appropriate response would be__?

A. Explain to the patient that you would be happy to treat his condition ,however since you haven't formally evaluated him it would be unfair to prescribe treatment over the phone .

B. Arrange a time for the patient to come into your clinic for immediate treatment .

C. Prescribe flexion exercises and ice every three hours .

D. Refer the patient to a qualified physician .

16. Which pulse site can be used to assess a patient in cardiac arrest and to monitor lower extremity circulation ?

A. Femoral .

B. Pedal

C. Popliteal

D. Radial .

17. The Q angle is designed as a measurement to determine the amount of lateral force on the patella . Which three bony landmarks are used to measure the Q angle?

A. Anterior superior iliac spine , superior border of the patella ,tibial tubercle .

B. **Anterior superior iliac ,midpoint of the patella ,tibial tubercle**.

C. Anterior superior iliac spine , inferior border of the patella ,midpoint of the patella tendon .

D. Greater trochanter , midpoint of the patella ,superior border of the patella tendon.

18. A 55 year old female diagnosed with a right hip intertrochanteric fracture is eight weeks status post open reduction and internal fixation with a plate and pinning . The patient currently has pain on hip flexion and abduction . Acceptable modalities for this patient would include all of the following except .

A. Hot packs .
B. Whirlpool
C. Pulsed ultrasound .
D. Shortwave diathermy .

19. Which of the following is not a realistic goal of a phase II cardiac rehabilitation program?

A. To increase exercise capacity and endurance .
B. To teach the patient self monitoring techniques .
C. To assess cardiovascular responses to work .
D. All of the above are realistic goals .

20. Why is a slight anterior pelvic tilt important when positioning a patient in a wheelchair?

A. **Allows for weight bearing on the ischial tuberosities** .
B. Provides flexion in the low back and extension in the hips .
C. Provides a stable base of support for control of the lower extremities
D. All of the above .

21.when isokinetically testing a patient which standard orthopedic testing procedure is not accurate ?

A. The patient must be informed and educated prior to testing .
B. **The involved side should be tested first to minimize the effects of muscle fatigue** .
C. Verbal commands should be consistent from one test to the next .
D. The patient should perform both submaximal and maximal warm-ups prior to testing .

22. when performing goniometric measurements a joint should be placed in the anatomical position . Which joint motions are exceptions to this rule ?

A. Forearm supination and pronation .
B. Hip rotation .
C. Shoulder rotation .
D. All of the above .

23. an order for chest physical therapy is received for an 82 year old female . The patient recently underwent surgery for a hip fracture and has been taking Coumadin postoperatively . She has a history of multiple compression fractures of thoracic vertebrae . The greatest amount of caution should be taken in the administration ___?

A. Diaphragmatic breathing exercises .

B. Postural drainage .

C. **Therapeutic percussion** .

D. Pursed lip breathing .

24. which technique would be the most appropriate when assessing the pulse of a patient after exercise ?

A. Placing the palm of the hand firmly over the pulse site .

B. Counting the pulse for five seconds and multiplying by twelve to determine the beats per minute .

C. **Selecting a pulse site that will not cause discomfort and subsequently alter the pulse rate** .

D. Selecting three to five separate pulse sites to assure an accurate measurement .

25. When a CVA patient lies on the hemiplegic side all of the following are correct except ____?

A. Spasticity is decreased due to elongation of the affected side .

B. **Awareness of the side is increased** .

C. The scapula should be in a retracted position .

D. The hemiplegic leg should be extended at the hip and slightly flexed at the knee .

26. while palpating the wrist and hand , it should become apparent that the hamate articulates with the ____metacarpal?

A. First .

B. Second.

C. Third.

D. Fourth.

27. A physical therapist evaluates a five year old child's gait . The therapist notes the child to be unsteady with a wide base of support . the child appears to lurch at times with minimal truncal bobbing in an anterior and posterior direction . The child can not maintain a standing position with the feet placed together for more than five seconds . The area of the brain most likely affected is the ____?

A. Corticospinal tracts .
B. Basal ganglion .
C. Substantia nigra .
D. **Cerebellar hemisphere** .

28. If the axillary nerve was severed, what muscle could laterally rotate the humerus?

A. Teres major .
B. Subscapularis .
C. Infraspinatus .
D. Trapezius .

29 . A group of graduate physical therapy students design a study to determine the effects of noise level on the ability to perform a physical skill . In their study noise is the ___?

A. Independent variable ?
B. Dependent variable .
C. Criterion variable .
D. Extraneous variable .

30. A physical therapist elects to begin joint mobilization by using distraction . Distraction is used for all of the following except______? .

A. **To unweight the joint surfaces** .
B. To relieve pressure on an intra – articular structure .
C. To stretch the joint capsule or adhesions .
D. To determine the nature of resistance at end range .

31. the single most important factor in an exercise program designed to increase muscular strength is ____?

A. The recovery time between exercise sets .
B. The number of repetitions per set .
C. The duration of the exercise session .
D. The intensity of the exercise .

32. while talking to a patient about their previous exercise program ,the patient mentions that he used to exclusively perform negative work . Negative work or a negative a negative muscular contraction is best defined as ____?

A. The muscle develops tension and increases in length .

B. The muscle develops tension and decreases in length .

C. Muscle length remains constant as tension is developed .

D. **Purposeful voluntary motion** .

33. A 45 year old obese woman in a long leg cast is attempting a sliding board transfer to a mat table , Which of the following instructions would be incurred ?

A. **Lean away from the mat and place the sliding board under your buttock**.

B. Perform a series of small pushups gradually moving closer to the mat .

C. Grasp the edge of the sliding board with your fingers to secure additional support .

D. Maintain a slight forward trunk position while transferring at all times .

34. A patient three months status post total knee referred to physical therapy for range of motion and strengthening exercises . Which treatment technique would be inappropriate for the patient ?

A. Active stretching using the contract – relax technique .

B. Joint mobilization to increase joint play .

C. **Exercise on a stationary bicycle against mild resistance** .

D. Performing straight raising , short arc extension , and knee flexion exercises using light weights

35. A physical therapist is evaluating a 43 year old female who has been diagnosed with a non – displaced fracture of the greater tuberosity of the humerus . The patient is unsure if she is supposed to keep her arm in the sling given to her by the physician. An appropriate course of action would be to ?

A. Instruct the patient to wear the sling at all times .

B. Instruct the patient not to use the sling because it will inhibit her range of motion .

C. Use your best judgment based on how the referring physician usually treats humerus fractures.

D. Contact the physician immediately and ask what instructions were given to the patient.

36. the most common site for an ulnar nerve injury is at the ___?

A. Brachial plexus.
B. Medial epicondyle of the humerus.
C. Superficial surface of the flexor retinaculum.
D. **Distal wrist crease.**

37. A 12 year old female became anoxic in a near drowning . The patient can perform dynamic activities in quadruped on a firm mat . The next posture to attain in the developmental sequence is ___?

A. Half kneeling .
B. **Tall kneeling** .
C. Plantigrade .
D. Standing .

38. Which type of electrical current would most likely select if your treatment objective is to stimulate denervated muscle ?

A. Alternating current .
B. **Direct current** .
C. Pulsatile current .
D. Denervated muscle responds in a similar manner to all types of current .

39. an 86 year old female is restricted to partial weight bearing on the left leg after a total hip replacement . Her upper extremity strength is 3\ 5 and she resides alone . Which assistive device would be the most appropriate for this patient ?

A. Lofstrand crutches .
B. **Axillary crutches** .
C. Large base quad cane .
D. Two wheel rolling walker .

40. which of the following situations would be contraindicated when using transcutaneous electrical nerve stimulation ?

A. Use over an arthritic joint .
B. Use during labor and delivery .
C. Use over a pregnant uterus.
D. **Use to diminish phantom limb sensation** .

41. An order is received to perform chest physical therapy on a patient who is status post abdominal surgery. A chart review identifies right atelectasis . During your first session the primary exercise that you will teach the patient is ___?

A. Reflex cough technique .
B. Codman pendulum exercises .
C. **Deep diaphragmatic breathing** .
D. Quick paced shallow breathing .

42. The position of a joint that is ideal for palpation is __?

A. Closed packed .
B. Open packed .
C. 90 degrees joint flexion .
D. Joint position does not matter .

43. Preoperative testing is performed prior to cardiac surgery in order to provide additional information on a patient's cardiopulmonary status . Which of the following diagnostic tests is of least value in the situation described above ?

A. Arterial blood gases .
B. Electrocardiogram .
C. Cardiac catheterization .
D. Magnetic resonance imaging .

44. There are a variety of factors which can significantly influence normal respiration including age ,sex ,stature and exercise . Which statement describing these is not accurate ?

A. The respiratory rate of a newborn is between 30 and 60 breaths per minute .
B. Men generally have a larger vital capacity than women .
C. Stout or obese subjects generally have a larger vital capacity than tall ,thin individuals .
D. Respiratory rate and depth will increase as a result of increased oxygen consumption and carbon dioxide production .

45. A physical therapist is treating a neurological patient with a flaccid left side . In order to facilitate muscular activity the treatment plan should include ___?

A. Weight bearing, tapping, elevation.
B. Vibration, tapping, prolonged stretch .
C. **Weight bearing , tapping ,approximation** .
D. Approximation, elevation ,prolonged stretch .

46. Developmentally, tonic stability is most evident when a child can___?

A. Lift their head while in prone .
B. Hold their head in neutral while sitting upright .
C. **Maintain pivot prone posture** .
D. None of the above .

47. a patient diagnosed with lateral epicondylitis is referred to physical therapy . The therapist elects to use iontophoresis over the lateral epicondyle area . Which type of current would the therapist probably use to administer the treatment ?

A. Direct.
B. Alternating.
C. Pulsatile.
D. Interferential.

48. Broca's area is affected by a CVA that occurs in which of the following vessels ?

A. **Anterior cerebral artery**.
B. Middle cerebral artery.
C. Posterior cerebral artery.
D. Basilar artery.

49. A patient was involved in a motor vehicle accident and sustained a fracture of the proximal fibula. The fracture injured the motor component of the common peroneal nerve. Ankle dorsiflexion is tested as 2\5. The most appropriate intervention to assist the patient with activities of daily living would be ___?

A. Electrical stimulation.
B. **An orthosis** .
C. An exercise program .
D. An aquatic program .

50. Communication and perceptual problems are extremely common in hemiplegic patients . Which clinical problem would be characteristic of a right CVA?

A. Inability to recognize symbols to do simple computations /
B. **Inability to plan and execute serial steps in activities** .
C. Distorted awareness and impression of self .
D. Lacks functional speech .

51. There are a variety of sizes of wheelchairs which can accommodate individuals of almost any size or build . Assuming a full grown adult of average size and build , which of the following wheelchair dimensions would be the most appropriate ?

A. Seat width 14 inches ,seat depth 12 inches .
B. Seat width 16 inches ,seat depth 14 inches .
C. Seat width 18 inches ,seat depth 16 inches .
D. Seat width 20 inches ,seat depth 18 inches .

52. a therapist is concerned about the possibility of a patient developing a deep vein thrombosis following surgery . Which of the following special tests would provide the therapist with information on a deep vein thrombosis ?

A. Bunnel - Littler test
B. Homans' sign .
C. Kleiger test .
D. Froment's sign .

53. An 81 year old female diagnosed with left – sided congestive heart failure is referred to physical therapy . Which objective finding is most commonly associated with left – sided heart failure ?

A. Distended neck veins .
B. **Liver and spleen congestion** .
C. Peripheral edema .
D. Pulmonary edema.

54. a patient tells her therapist she been ill for the past week and feels it may have contributed to her poor performance in selected exercise activities . The therapist attempts to document the patient's inability to achieve weekly goals with the following entry : " patient may have performed poorly during the last week secondary to recent illness " . This entry would belong in the ___________ section of a SOAP note ?

A. Subjective.
B. Objective.
C. Assessment.
D. Plan.

55. A patient in a phase l cardiac rehabilitation program begins walking with assistance. Which of the following monitoring techniques would not be necessary in a phase 1 program ?

A. Blood pressure.
B. Electrocardiography.
C. Electromyography.
D. Heart rate .

56. a therapist completes a developmental assessment on a five month old infant .

If the therapist elects to evaluate the infant's palmar grasp reflex , which of the following stimuli is the most appropriate ?

A. Contact to the ball of the foot in upright standing .
B. Maintained pressure to the palm of the hand .
C. Noxious stimulus to the palm of the hand
D. Sudden change in the position of the hand.

57. An effective manager will rely on well established policies and procedures when disciplining an employee. Which of the following actions would be considered inappropriate?

A. Begin with a clear description of the unacceptable behavior .
B. **Indicate what action should be taken by the employee** .
C. Set a follow up date to discuss the situation .
D. Focus the discussion on the individual and not on a performance discrepancy .

58. A physician places a patient with moderate hypertension on diuretics . What is the most serious side effect of diuretics?

A. Fluid depletion and electrolyte imbalance .
B. Gastrointestinal disturbances .
C. Hypokalemia .
D. Weakness and fatigue .

59. Lontophoresis and phonophoresis are commonly used techniques of delivering a specified medication through the skin to underlying tissue . Which route of administration dose inotophoresis and phonophorsis utilize ?

A. Inhalation .
B. Sublingual .
C. Topical .
D. Transdermal .

60. changes in the health care delivery system have significantly influenced physical therapy practice patterns . Which of the following trends is probably not reflective of practice patterns over the last decade ?

A. An increased percentage of therapists of therapists working primarily in hospitals .

B. An increased percentage of therapists of therapists working.

C. An increased percentage of therapists of therapists working in home care .

D. An increased percentage of therapists of therapists working in private practice settings .

61. a therapist reviews the medical chart of a cardiac patient with a history of recurrent serous dysrhythmias . The therapist is concerned about the patient's past medical history and would like to monitor the patient during all activities . Which of the following parameters would be the most valuable for the therapist to monitor ?

A. **Heart rate** .
B. Electrocardiogram .
C. Blood pressure .
D. Respiratory rate .

62. a patient with a lengthy history of substance abuse is referred to physical therapy after sustaining multiple injuries in a motor vehicle accident . Which of the following controlled substance dose not foster physical dependence ?

A. Depressants .
B. Hallucinogens .
C. Narcotics.
D. Stimulants .

63. gloves can be an effective means of protective asepsis ; what type of isolation requires the use of gloves by all individuals entering the room ?

A. AFB .
B. Contact
C. Respiratory .
D. Strict .

64.a group of physical therapy statements a research project entitled , " Effects of Ultrasound on Blood Flow and Nerve Conduction Velocity " . the dependent variable in the student's study is ___?

A. Ultrasound.
B. Ultrasound and blood flow .
C. Ultrasound and nerve conduction velocity .
D. Blood flow and nerve conduction velocity .

65. a therapist instructs a 65 year old female in ambulation activities . The therapist instructs the patient to place their affected foot on the ground but to avoid transmitting weight through the foot. The therapist's instructions best describe ___weight bearing ?

A. Non .
B. Touch down .
C. Partial .
D. Full .

66. A therapist instructs a patient to close their eyes and hold out their hand. The therapist places a series of different weights in the palients hand one at a time. The patient is then asked to identify the comparative weight of the objects. This method of sensory testing is used to evaluate ____?

A. Barognosis.
B. Graphesthesia.
C. Recognition of texture.
D. Stereognosis.

67. A therapist requests nursing to take the temperature of a patient of a patient before their physical therapy session. Which of the following temperatures would be classified as normal?

A. **35 degrees Celsius** .
B. 36 degrees Celsius.
C. 37 degrees Celsius.
D. 38 degrees Celsius.

68. Secretion removal techniques are an integral component of a pulmonary rehabilitation program. What is the simplest method to clear secretions from the upper airways ?

A. Cough .
B. Percussion .
C. **Shaking** .
D. Vibration .

69. A therapist develops a list of goals for a developing phase 1 cardiac rehabilitation program . which of the following goals would not be appropriate for a phase 1 program?

A. Initiate. Low levels of activity.
B. Educate patients and their families .
C. Measure the effectiveness of medication in controlling the patient's cardiovascular status during activity .
D. Retum the patient to safe vocational and recreational activities .

70. A twelve year old boy sitting in the physical therapy waiting area suddenly grasps his throat and appears to be in distress . The patient slowly stands but is obviously unable to breathe . The therapist recognizing the signs of an airway obstruction should administer ______?

A. Abdominal thrusts .
B. Chest thrusts .
C. Back blows .
D. Back blows in combination with abdominal thrusts

71. A therapist performs girth measurements on a patient rehabilitating from knee surgery .

The therapist takes the measurements 5 cm and 10 cm above the superior pole of the patella with the patient in supine . The girth measurements are recorded as 32 cm and 37 cm on the right and 34 cm and 40 cm on the left . Which of the following conclusions can be formed on the strength of the patient's quadriceps ?

A. The right quadriceps will be capable of producing a greater force than the left .
B. The left quadriceps will be capable of producing a greater force than the right .
C. The right and left quadriceps will be capable of producing equal force .
D. Not enough information is given to form a conclusion .

72. A patient receiving physical therapy treatment complains to their therapist that they are experiencing a great deal of pain around their I.V. site . The therapists most appropriate response would be to ____?

A. Reposition the I.V.
B. Remove the I.V.
C. Turn off the I.V.
D. Contact nursing .

73. A therapist examines an athlete immediately after sustaining a knee injury while playing softball. The therapist evaluates the filamentous integrity of the athlete's knee by performing series of special tests. The therapist concludes that during an anterior draw test the patient's knee demonstrates excessive anterior translation of the tibia on the femur. An injury to what structure other than the anterior cruciate ligament could give the impression of excessive anterior instability?

A. Lateral collateral ligament .
B. Medial collateral ligament .
C. Medial meniscus .
D. **Posterior cruciate ligament** .

74. A physical therapist instructs a woman with left unilateral lower extremity weakness in descending stairs . The most appropriate position for the therapist to guard the patient is _____?

A. In front of the patient toward their left side .
B. In front of the patient toward their right side .
C. Behind the patient toward their left side .
D. Behind the patient toward their right side .

75. A therapist completes a sensory evaluation on an incomplete T7-T8 paraplegic . The therapist evaluates the patient's sensation using a piece of cotton . The therapist applies the cotton in a random fashion and the patient is asked to indicate when they feel the stimulus. This method of sensory testing used to evaluate _____?

A. Kinesthesia.
B. Light touch.
C. Proprioception.
D. Superficial pain.

76. A therapist is growing increasingly concerned about a patient that is demonstrating symptoms that are consistent with neoplastic activity . what is the most significant symptom a rapidly growing neoplasm ?

A. Fatigue .
B. **Swelling** .
C. Tenderness to palpation .
D. Pain .

77.a therapist instructs a patient to expire maximally after taking a maximal inspiration . The therapist can use these instructions to measure the patient's __?

A. Expiratory reserve volume .
B. Aspiratory reserve volume .
C. Total lung capacity .
D. **Vital capacity** .

78. a six month ,twenty- two pound baby girl fractures the shaft of her left femur . What type of traction is the most common to treat the baby's condition ?

A. Bryant's traction .
B. Buck's traction .
C. Lateral skeletal traction .
D. **Russell's traction** .

79. chest percussion and vibration are appropriate bronchial drainage techniques for all of the following except the ?

A. **Anterior apical segment** .
B. Lingual .
C. Left middle lobe .
D. Right middle lobe .

80. a patient rehabilitating from a motor vehicle accident complains of pain along their chest wall during inspiration . The therapist attempts to identify the cause of the pain by palpating along the chest wall. While palpating the therapist detects grating with associated point tenderness on a small portion of the chest wall . Which type of injury could best explain the patient's symptoms ?

A. Rib fracture .
B. Subluxation of costal cartilage .
C. **Intercostals fibrositis** .
D. Pulmonary embolus .

81.a therapist instructs a patient in diaphragmatic breathing exercises . Which of the following instructions would not be helpful in teaching diaphragmatic breathing ?

A. Place your dominant hand over your mid rectus area .
B. Place your dominant hand over your mid sternal area.
C. Inspire slowly through your nose .
D. As you expire watch your lover hand rise .

82. the role of the physical therapist assistant varies widely depending on the individual department policy and the state practice act . Which of the following duties is generally not performed by a physical therapist assistant ?
A. Adjusting ambulation equipment .
B. Applying modalities .
C. Documentation including evaluations and discharge summaries .
D. Wound care .

83. a therapist volunteers to assist participants at the finish line in a 10k road race . The race takes place on a hot and humid day and some of the race organizers are concerned about the potential for heat related disorders such as heat exhaustion and heat stroke .
The most significant variables to differentiate between heat exhaustion and heat stroke are _____?
A. Blood pressure and pulse rate .
B. Coordination and level of fatigue .
C. **Pulse rate and skin temperature** .
D. Pupil dilation and blood pressure .

84. a therapist instructs a patient on how to rise from a chair before beginning ambulation activities with a walker . Which of the following instructions would be helpful to the patient ?
A. Place both hands on the walker and pull yourself to a standing position .
B. Push up on the chair with one hand and place the other hand on the edge of the walker for balance .
C. **Push up on the chair with both hands and reach for the walker once you are standing** .
D. Push up on the chair with both hands and reach for the walker while rising .

85. a physical therapist palpates proximally along the lateral border of the fifth metatarsal of a patient's foot . Which bone would be palpable as the therapist continues to palpate proximally along the lateral border of the foot ?

A. Cuboid .

B. Second cuneiform .

C. **Third cuneiform** .

D. Navicular .

86. a consistent testing procedure is essential to obtain valid data when performing a sensory evaluation . Which of the following guidelines is not accurate when administering a sensory evaluation ?

A. A demonstration of each procedure should be performed before the actual test .

B. Testing should be carried out following the main sensory nerves and their corresponding dermatomes .

C. The discriminative sensations are tested before the protective sensations .

D. Testing of the extremities is usually performed in a distal to proximal progression .

87. a child fractures their femoral shaft after falling out of a tree house . the most serious complication of a femoral shaft fractures is ___?

A. Avascular necrosis .

B. Coax vara deformity .

C. **Persistent knee stiffness** .

D. Volkmann's ischemia .

88.a therapist prepares to treat a patient with limited ankle range of motion using joint mobilization techniques . If the therapist elects to mobilize the talocrural joint in the loose packed position, the therapist should position the ankle in ____?

A. 10 degrees dorsiflexion, 5degrees inversion.

B. 10 degrees dorsiflexion, 5degrees eversion.

C. 10 degrees plantarflexion, 5degrees inversion.

D. 10 degrees plantarflexion, 5degrees eversion.

89. a therapist records the vital signs of a 45 year old male prior to beginning physical therapy treatment . The therapist palpates the patient's radial pulse for 15 seconds but has difficulty computing the actual pulse rate since the rhythm is irregular. The most appropriate method to identify the actual pulse rate is to ___?

A. Recheck hand positioning and palpate the radial pulse for an additional 15 seconds .

B. Select another pulse site and palate for 15 seconds .

C. **Palpate the radial pulse for a full minute** .

D. Record the original pulse rate and document the original rhythm .

90. health care facilities utilize a variety of employees including physical therapists ,physical therapist assistants and physical therapy aides . Which activity would be inappropriate for a physical therapy aide ?

A. Assist patients in preparation for treatment .

B. Make daily entries in the patient's medical record .

C. Assist patients in the safe practice of exercise activities.

D. Perform specific treatment procedures predetermined for each patient by the physical therapist .

91. A patient rehabilitating from a lower extremity injury is referred to physical therapy for hydrotherapy treatments. The therapist would like the patient to fully extend their involved lower extremity while in the hydrotherapy tank. Which type of whirlpool would not allow the patient to extend their involved lower extremity?

A. Hubbard tank .

B. **Highboy tank** .

C. Lowboy tank .

D. Walk tank .

92.A patient recovering in the hospital from a total knee replacement is evaluated by a physical therapist . Assuming an uncomplicated recovery , what knee range of motion would be a realistic goal for the patient prior to discharge from the hospital ?

A. 0-60 degrees .

B. 0-90 degrees .

C. 15-90 degrees .

D. 15-105 degrees .

93. a therapist orders a wheelchair for a T8-T9 paraplegic . which wheelchair option is not usually necessary for a paraplegic ?

A. Detachable leg rests .

B. Pneumatic tires .

C. Removable arms .

D. Wheel rim projections .

94. a therapist conducts an exercise test on a pulmonary patient to determine the amount of impairment caused by their present illness . which of the following would not be considered acceptable criteria for stopping a pulmonary exercise test ?

A. Cardiac ischemia or arrhythmias .

B. Increased cardiac output .

C. Maximal shortness of breath .

D. Symptoms of fatigue .

95. A therapist instructs a patient ambulating with axillary crutches to move cautiously and deliberately on any potentially hazardous surface . What gait pattern would allow the patient maximum security ?

A. Swing to .

B. **Swing through** .

C. Three point .

D. Four point .

96. a therapist completes a physical examination of a patient before beginning pulmonary rehabilitation .if the therapist would like to obtain information on the use of accessory muscles during ventilation ,what area should the therapist observe ?

A. Abdomen .

B. Mid thorax .

C. Neck and shoulders .

D. **Shoulders and upper thorax** .

97. A therapist performs a manual muscle test on a patient with unilateral upper extremity weakness . The patient is able to complete 75 % of the available range of motion with gravity eliminated . The therapist should record the muscle grade as ____?

A. Poor plus .

B. Poor .

C. Poor minus .

D. Trace plus .

98. a physical therapist performs a manual muscle test on a patient's hip flexors . The therapist attempts to complete the test in supine but the patient has difficulty holding the limb in the test position. What alternate test position would be appropriate to test the patient's hip flexors?

A. Prone .
B. Sidelying .
C. sitting .
D. Standing.

99. a therapist measures a patient for a straight cane prior to beginning ambulation activities which gross measurement method would generally provide the patient with the appropriate size assistive device?

A. Measuring from the fibula head straight to the floor and multiplying by two .
B. Measuring from the iliac rest straight to the floor.
C. Measuring from the greater trochanter to the floor.
D. Dividing the patient's height by two and adding three inches .

100. if a goal for a hemiplegic patient is to transfer form supine to standing ,what must they be able to do first ?

A. Ambulate with a walker .
B. Independently propel a wheelchair .
C. Roll.
D. assume quadruped .

101. a manual muscle test is conducted on the medial portion of the deltoid . the muscle is able to move through the full range of motion with gravity eliminated ,but can not function against gravity . The results of the manual muscle test should be recorded as ____?

A. Trace .
B. Poor.
C. Fair.
D. Good.

102. Which treatment should be included in a physical therapy program for a patient with atherosclerosis ?

A. Ambulation .
B. **Reflex heating** .
C. Gentle active exercise .
D. All of the above .

103. a female tennis player is referred to physical therapy diagnosed with medial tibial stress syndrome . the athlete mentions that her shins seem to ache all of the time and the pain seeps to intensify with activity .palpation reveals severs point tenderness along the medial posterior edge of the tibia . further evaluation reveals markedly pronated feet X- rays show no indication of a stress fracture .

The most appropriate exercise to begin immediately is _____?

A. Static stretching of the Achilles tendon and anterior portion of the lower leg .
B. Light jogging on flat terrain .
C. Progressive resistive exercise in both planter and dorsiflexion /
D. **Plantarflexion resistive exercise** .

104. Therapeutic activities based on the above situation would include all of the following except ___?

A. **Low dye taping applied to the arch to correct pronation with weight bearing** .
B. Ice packs or ice massage followed by stretching .
C. A heel lift to limit excessive stress on the medial tibia .
D. Ultrasound at 0.5- 0.75 w/cm2 for 7 minutes

105 . a physical therapist uses rhythmic initiation to assist a patient in learning to roll from supine to prone . The physical therapist's first verbal command should be ___?

A. " slowly roll over by yourself ".
B. "help me roll you over " .
C. "**stop me from rolling you over** " .
D. "Relax and let me move you" .

106. a physical therapist begins to examine a 22 year old soccer player who suffered an inversion ankle sprain 24 hours ago . the therapist immediately notices the athlete's right ankle is moderately edematous . What valuable information has the therapist already acquired?

A. The athlete will probably be out of competitive soccer for at least two weeks because of the severity of the sprain .
B. The athlete should be non – weight bearing at all times due to the amount of effusion in the ankle .
C. The sprain is a third degree ligament sprain although no timetable for recovery is available based on the information given .
D. The therapist still does have an accurate indication to the severity of the injury .

107. which clinical finding would you expect to see from a patient diagnosed with right sided heart failure ?

A. Increased stoke volume .

B. Pulmonary edema.

C. Edematous ankles.

D. Bradycardia .

108. which statement best describes a traumatic brain injured patient who is at level Vll , " Automatic – Appropriate " , of the Ranchos Los Amigos levels of cognitive functioning ?

A. The patient is in a heightened state of activity with a severe impairment in the ability to process information.

B. The patient has gross attention to the environment, is highly distractible, and lacks ability to focus on a task without redirection .

C. The patient is able to go through their daily routine in a robot like fashion with little to no confusion and has an increased awareness of the environment .

D. The patient shows goal directed behavior but relies on external input for direction .

109. which statement would not be considered good advice for a patient following a total hip replacement ?

A. When turning ,pivot to the affected side .

B. Do not cross your legs .

C. **Avoid low chairs** .

D. Keep a pillow between your legs while sleeping .

110.a physical therapist uses proprioceptive neuromuscular facilitation to increase joint range of motion using the hold – relax technique . The technique utilizes an ___contraction , which is used at the end point of the available range of motion ?

A. Isotonic .

B. Isometric .

C. Isokinetic.

D. Eccentric .

111. based on the "rule of nines " ,an adult who has burns on their anterior right arm ,the anterior portion of the thorax and the genital region would be classified as having burns over ____ percent of their body ?

A. 19

B. 22.5

C. 23.5

D. 28

112. When a patient is lying supine and the surface suddenly tilts laterally to the left ,the balance reaction that occurs is _____?

A. The neck flexes laterally to the left ,trunk side flexion to the left ,arm and leg on the left abduct and extend .

B. The neck flexes ,trunk side flexion to the left ,arm and leg on the left abduct and extend .

C. The neck flexes laterally to the right ,trunk side flexion to the left ,arm and leg on the right abduct and extend .

D. The neck flexes laterally to the right ,trunk side flexion to the right ,arm and leg on the right abduct and extend .

113. Which general rule best determines the length of treatment when using ultrasound ?

A. 2 minutes for every area that is 2-3 times the size of the transducer face .

B. 5 minutes for every area that is 2-3 times the size of the transducer face .

C. 5 minutes is the maximum treatment time regardless of the treatment area .

D. 10 minutes is the maximum treatment time regardless of the treatment area.

114. Electrical stimulating currents may produce the following effects:

a. muscle contraction

b. net ion movement

c. decrease in pain

d. all of the above

115. A patient is referred to physical therapy for range of motion and strengthening exercises after sustaining an anterior dislocation of the right shoulder . The patient was immobilized in a sling and swathe for four weeks and it has been five weeks since the injury date. Which activity would be the most appropriate for the patient to begin strengthening of the involved shoulder ?

A. Internal and external rotation in Sidelying with two pound weights within the available range of motion .

B. Internal and external rotation using elastic tubing with the arm at the patient's side and elbow at 90 degrees .

C. Isokinetic internal and external rotation at speeds less than 120 degrees per second .

D. Isometric strengthening exercises with the arm positioned at the patient's side.

116. which of the following developmental landmarks is usually not observed when evaluating a four month old infant ?

A. Eyes follow objects 180 degrees .

B. Infant laughs .

C. Transfers blocks from one hand to the other .

D. **Holds head erect when held in a sitting position**.

117. Which of the following factors is of least importance when selecting an assistive device ?

A. The patent's level of understanding .

B. The patient's height and weight .

C. The patient's upper and lower extremity strength .

D. The patient's level of coordination .

118. An above knee amputee is referred to physical for gait training. During the initial evaluation the therapist identifies trunk bending toward the affected side and a pelvic drop over the prosthetic leg during the stance phase of gait. The therapist may conclude _____?

A. The prosthesis is too long.

B. **The prosthesis is too short** .

C. The socket diameter is too small .

D. There is an increased amount of edema in the residual limb .

119. A therapist passively moves a patient's upper extremity through a selected range of motion with the patient's eyes closed . The patient is then asked to describe verbally the direction and range of movement of the upper extremity . This technique can be used to evaluate _____?

A. Barognosis
B. Graphesthesia.
C. Kinesthesia.
D. Stereognosis.

120. Policies and procedures are used to carry out effective and efficient management. Which statement describing policies and procedures is not accurate?

A. Policies should be written on topics such as dress code , promotions and discipline .
B. Policies should be established on an oral basis and subjected to individual interpretation.
C. Policies should be reviewed on a regular basis and revised if indicated .
D. Policies should be assembled into a manual for quick and easy reference

121. a forced expiratory volume in one second that is less than 80 % of the vital capacity would be indicative _____?

A. Obstructive pulmonary disease .
B. **Restrictive pulmonary disease** .
C. Obstructive and restrictive pulmonary disease .
D. None of the above .

122. Treatment for a child diagnosed with cerebral palsy must be individualized to each child's needs . In planning a treatment program that will remain effective over time ,all of the following guidelines should be incorporated except ___?

A. Treatment goals should be designed to meet specific problems .
B. Play should not be integrated into therapy as it distracts from direct movement responses .
C. Therapy should encourage active responses from the child .
D. Repetition is an important component of motor learning.

123. a therapist records grip strength measurements on a patient diagnosed with bilateral carpal tunnel syndrome . Which description does not accurately describe typical results when using a hand held dynamometer?

A. A bell curve is seen when charting multiple recordings .

B. **Twenty to twenty five percent differences in grip strength may be seen between the dominant and nondominant hands** .

C. Discrepancies of more than twenty five percent in a test – retest situation indicate the patient is not exerting maximal force .

D. An individual who does not exert maximal force for each test will not show the typical bell curve .

124. Iontophoresis is used to treat all of the following conditions except____?

A. Muscle spasms .

B. Ischemic skin ulcers .

C. Fungal infection .

D. Bursitis and tendonitis .

125. Bandaging a residual limb is one method of postoperative dressing . the purpose of bandaging includes all of the following except _______?

A. Provides complete protection against accidental bruising and bumping .

B. Reduces edema .

C. Supports the surgical wound .

D. Aids in preventing contractures .

126. Which of the following relaxation techniques would be beneficial during labor and delivery?

A. Attention focusing .

B. Systematic relaxation .

C. Imagery .

D. All of the above are beneficial .

127. a frequently administered treatment for cancer in which high doses of energy are administered to the body to stop cell replication is termed___?

A. Immunotherapy.

B. Radiation therapy.

C. Chemotherapy.

D. **Surgical intervention**.

128. The intermittent use of tone reducing lower extremity bivalve casting for children with cerebral palsy hypothesizes that casting may____?

A. Increase compensatory stabilizing efforts
B. Cause flexing of the toes which inhibits the plantar grasp response.
C. Facilitate trunk stability by reducing contractures of the foot .
D. Inhibit the use of extensor thrust by preventing dorsiflexion.

129. which injury mechanism is the most common for spinal cord injury in the cervical spine ?
A. Cervical hyperextension.
B. Cervical hyperflexion.
C. Cervical rotation.
D. Cervical lateral flexion .

130. A patient is status post radical mastectomy secondary to breast cancer .physical therapy intervention includes____?

A. Monitoring postoperative circumferential measurements .
B. **Passive and active assistive range of motion** .
C. Elevation and positioning of the upper extremity .
D. All of the above .

131.A person with an anterior cruciate ligament insufficiency , who elects not to have surgical reconstruction ,could probably expect to reach which minimal functional level ?
A. Able to participate in all sports .
B. Able to participate in light recreational sports .
C. Can not play any type of sport.
D. Problems with normal walking .

132. A pregnant woman must closely monitor her heart rate during exercise . A safe level of exertion is calculated by a maximum heart rate of _____?
A. 200-mother's age .
B. 60%-70%of (220-mother's age) .
C. 30 %of (220 - mother's age) .
D. Maximum heart rate of 210 .

133. In determining guidelines for pregnancy and postpartum exercise programs , which of the following factors should be considered?

A. Deep flexion and extension of joints should be avoided .
B. **Heart rate should be measured at times of peak activity** .
C. Liquids should be taken before and after exercise to prevent dehydration .
D. All are appropriate guidelines .

134. all of the following modalities would be indicated in the treatment of a pregnant woman except_____?
A. Vapocoolant sprays in the treatment of muscle spasms .
B. Transcutaneous electrical nerve stimulation used for pain relief during labor .
C. Phonophoresis in the treatment of lateral epicondylitis.
D. All of the above are indicated during pregnancy.

135. a physical therapist performs a manual muscle test on the primary hip abductor . The therapist should perform the test while palpating the____?

A. Rectus femoris .
B. Gluteus medius .
C. Sartorius .
D. **Gluteus minimus** .

136. a manual muscle test of the triceps reveals evidence of alight contractility without any active range of motion . The muscle should be graded as ____?

A. Zero.
B. Trace.
C. Poor.
D. Fair.

137. Which of the following exercises is indicated for coordination training ?
A. Buerger - Allen exercises .
B. **Codman's exercises**.
C. Delorme's exercises.
D. Frenkel's exercises .

138.A 36 year old female is limited to 30 degrees of external rotation at the right shoulder . Which shoulder mobilization technique might prove to be beneficial ?

A. Anterior glide of the head of the humerus .
B. Posterior glide of the head of the humerus .
C. Inferior glide of the humeral head on the glenoid .
D. Superior glide of the humeral head on the glenoid.

139. what objective finding would you not expect to see while observing the gait of a patient with an acute grade II lateral ligament injury ?

A. Decreased step length .
B. Decreased cadence .
C. Decreased toe off on the involved lower extremity .
D. Decreased base of support .

140. a physical therapist should adhere to all of the following instructions when rolling a patient on a floor mat except ____?

A. Position yourself away from the side the patient is going to be turned toward.
B. Assume a half –kneeling position with the "down " knee at the level of the patient's hips .
C. Move with the patient in order to allow the patient to complete a full roll .
D. Communicate with the patient prior to any change in position

141. whirlpool treatments are commonly used in physical therapy to stimulate circulation , promote muscle relaxation and provide pain relief . which statement regarding hydrotherapy is not true ?

A. **41-46 degrees Celsius are acceptable water temperatures** .
B. The region to be treated should be thoroughly inspected prior to beginning treatment .
C. The tank and turbine should be cleaned after each treatment .
D. The agitation force should be adjusted initially at a minimum level and increased as desired .

142. A physical therapist observes a patient ambulating with a normal gait pattern. The therapist notices that approximately ____ percent of the normal gait cycle is spent in the stance phase ?

A. 20
B. 40
C. 60
D. 80

143. Which progressive resistive exercise would function to strengthen the infraspinatus and teres minor .

A. Extension of the shoulder with dumbbell weights .
B. Flexion of the shoulder with dumbbell weights .
C. **External rotation of the shoulder with elastic tubing** .
D. Internal rotation of the shoulder with elastic tubing .

144. a physical therapist palpates form a spinous process laterally along the second rib . What bony structure would expect the therapist to next encounter ?

A. Lateral border of the scapula .
B. Spine of the scapula .
C. **superior angle of the scapula** .
D. Acromion .

145. a physical therapist transfers a patient in a wheelchair down a curb with a forward approach . Which of the following instructions is correct ?

A. Have the patient lean forward .
B. Have the wheelchair brakes locked .
C. **Tilt the wheelchair backwards** .
D. Position yourself in front of the patient .

146. a physical therapist is attempting to passively stretch a patient's rectus femoris . The most effective position to stretch this muscle is ____?

A. Patient standing holding flexed knee .
B. Patient in prone with the therapist flexing the knee.
C. **Patient in sidelying with the therapist flexing the knee** .
D. Patient in prone with the therapist stabilizing the pelvis and flexing the knee .

147. a patient who is partial weight bearing on the left is ambulation with a walker . The patient advances the walker followed by a step with the left and then a step with the right . This gait pattern is termed ____?

A. Two-point .
B. Three- point .
C. Four – point ..
D. **Swing to** .

148. a physical therapist is treating a patient with a fractured left hip . the patient is weight bearing as tolerated and uses a large base quad cane for gait activities . Correct use of the quad cane would include ____?

A. Using the quad cane on the left with the longer legs positioned away from the patient .
B. Using the quad cane on the left with the longer legs positioned away from the patient .
C. Using the quad cane on the left with the longer legs positioned toward from the patient .
D. Using the quad cane on the left with the longer legs positioned toward from the patient .

149. a physical therapist observes a patient ambulating with a trendelenburg gait pattern . The deviation is caused by weakness of the _____?

A. Gluteus maximus .
B. Gluteus medius.
C. Gluteus minimus .
D. Piriformis .

150.a physical therapist reviews an initial evaluation of a patient diagnosed with an upper motor neuron disease . The evaluation documente the existence of a positive Babinski reflex . A positive Babinski reflex is characterized ___?

A. Extension of the great toe .
B. **Flexion the great toe** .
C. Abduction of the great toe .
D. Adduction of the great toe .

151. a therapist measures passive forearm pronation and concludes the results are within normal limits . Which measurement would be classified as within normal limits?

A. 65 degrees .
B. 85 degrees .
C. 105degrees .
D. 125 degrees .

152. A physical therapist obtains a goniometric measurement with a patient in supine . The stationary arm of the goniometer is positioned at the mid – axillary line of the trunk . The moveable arm is positioned along the humerus . this positioning of the goniometer can be used to measure ?

A. Shoulder flexion .
B. Internal and external rotation of the shoulder .
C. Shoulder extension.
D. Elbow extension .

153. a physical therapist is treating a stroke patient with a hypotonic left upper extremity . While performing sitting activities the position of choice for the left upper extremity is ____?

A. In the patient's lap .
B. **In a sling**.
C. Weight bearing through the upper extremity with the elbow extended .
D. Weight bearing through the upper extremity with the elbow and wrist flexed .

154. a physical therapist is treating a patient with lower extremity hypertonicity . All of the following could be used by the therapist to temporarily reduce tone except _____?

A. Gentle rocking .
B. Cryotherapy.
C. Approximation.
D. Sustained stretch .

155. a patient who recently sustained a gastrocnemies strain asks their therapies what factors may have contributed to the muscle strain . Which statement dose not accurately describe contributing factors to a muscle rupture or strain ?

A. The muscle may have been poorly prepared due to inadequate warm up .
B. The muscle may have been weakened by a previous injury .
C. **The muscle may have been subjected to prolonged heat exposure which would cause the muscle to be less contractile than normal** .
D. The muscle may previously have been extensively injured with resultant scar tissue formation.

156. Effective measures to prevent decubitus ulcers include all of the following except _______?
A. Frequent inspection of the skin for redness and signs of skin breakdown .
B. Turning and position changes every 12 hours .
C. **Early ambulation when possible** .
D. Passive range of motion exercises .

157. a physical therapist is consistently falling behind with their documentation due to an excessive patient load . The most appropriate response is to ___?
A. Discuss the situation with other staff physical therapists.
B. **Ignore the situation and attempt to complete the documentation in a timely fashion**.
C. Discuss the situation with their immediate supervisor.
D. Discuss the situation with the director of rehabilitation.

158. a physical therapist attempts to measure a patient's shoulder external rotation . proper positioning of the upper extremity in supine would best be described by which of the following ?
A. Shoulder abducted to 90 degrees , elbow flexed to 90 degrees .
B. Shoulder abducted to 90 degrees , elbow fully extended .
C. Shoulder abducted to 90 degrees , elbow flexed to 45 degrees.
D. Shoulder in neutral ,elbow flexed to 90 degrees .

159. a physical therapist observes a patient performing active hip abduction in supine . The patient in limited by 10 degrees in active hip abduction but appears to be moving through the full range of motion . What compensatory measures might the patient use to seemingly increase their hip abduction ?

A. **Hip flexion and external rotation** .
B. Hip flexion and internal rotation.
C. Hip hyperextension and external rotation.
D. Hip hyperextension and internal rotation.

160. According to standards of Ethical for physical therapists as written by the American physical therapy Association all of the following are correct except ___?

A. Physical therapists shall be responsive to and mutually supportive of colleagues and associates.
B. Physical therapists shall support research activities that contribute knowledge for improved patient care.
C. Physical therapists may not be remunerated for endorsement or advertisement of equipment or services to the lay public , physical therapists or other health professionals .
D. Physical therapists may enter into agreements with organizations to provide physical therapy services if such agreements do not violate the ethical principles of the association .

161. What massage technique is commonly used at the beginning and conclusion of treatment and is used during transitional periods prior to changing massage techniques?
A. Effleurage .
B. Petrissage .
C. Tapotement.
D. Vibration .

162. a 25 year old male is referred to physical therapy with limited right knee flexion . After evaluating the involved knee you feel the restriction is in a noncapsular pattern . Which of the following examples is not typical of a noncapsular restriction?
A. **Internal derangement**.
B. Arthritis, degenerative joint disease .
C. Isolated ligamentous adhesion .
D. Extra- articular tissue tightness .

163. restrictive lung diseases limit to varying degrees the maximum volumes of air can be inhaled and exhaled . Examples of restrictive diseases include all of the following except ___?

A. Emphysema .
B. A paralysis of respiratory muscles .
C. Pneumonia .
D. A bony abnormality of the chest .

164. a patient with a fractured right tibia who is partial weight bearing is referred to physical therapy for fitting and instruction with an assistive device . During the evaluation the physical therapist identifies poor lower extremity stability and coordination . Which assistive device would be most appropriate for the patient ?

A. **Axillary crutches** .
B. Cane .
C. Lofstrand crutches .
D. Walker .

165. Which of the following would probably not increase a patient's compliance with their home exercise program ?

A. The patient should receive both written and verbal instructions .
B. The program should be goal oriented .
C. The program should be individualized to the needs of the patient .
D. The program should take in excess of thirty minutes to successfully complete.

166. a therapist observes the gait of a right CVA patient with a metal upright ankle foot outhouses . The therapist notes left knee instability with left stance . The ankle foot orthosis most likely has___?

A. A long foot plate .
B. Inadequate knee lock .
C. Inadequate dorsiflexion stop .
D. **Inadequate dorsiflexion stop** .

167. a physical therapist is treating a patient with a limited straight leg raise of 40 degrees due to inadequate hamstring length . Which proprioceptive neuromuscular technique would be most appropriate to increase the patient's hamstring length?

A. Contract – relax .
B. Rhythmic initiation
C. **Rhythmic stabilization .**
D. Rhythmic rotation .

168. a short term goal for a CVA patient is to have patient perform a stand pivot transfer from abed to a wheelchair . This goal should be recorded in the ____section of a SOAP note ?

A. Subjective .
B. Objective .
C. Assessment .
D. Plan .

169. a physical therapy student practices assessing joint end – feel . The student would most accurately classify normal elbow extension end – feel as ___?

A. Boggy .
B. Bony .
C. Capsular .
D. Muscular .

170. A physical therapist observes a patient dragging their toe during the swing phase of gait . All of the following are possible causes except ___?

A. Weakness of dorsiflexors.
B. Spasticity of plantarflexors .
C. Weakness of plantarflexors.
D. Inadequate knee flexion.

171. A physical therapist notices a student physical therapist disrupting an outpatient's therapy session. The student continues to sit next to the patient and talk about unrelated issues.

A. discuss the situation with the student's clinical instructor
B. leave the student an anonymous message regarding their behavior
C. **confront the student immediately**
D. ignore the situation since you are not the clinical instructor

172. A physical therapist instructs a patient diagnosed with Parkinson's disease in ambulation activates. Which of the following might the therapist use to improve the quality of the patient's gait?

A**. decreased stride length**

B. increased trunk rotation

C. increased forward head posture

D. decreased base of support

173. When wrapping the residual limb of a below knee amputee a therapist should adhere to all of the following except_______?

A. the bandage should provide increased pressure proximally

B. **the bandage should be smooth and wrinkle free**

C. the bandage should be applied using diagonal turns

D. the bandage should be applied to encourage proximal joint extension

174. A physical therapist has a CVA patient weight bear through their affected upper extremity in a plantigrade position. The most appropriate hand position on the plinth is________?

A. **total palmar contact with the fingers fully extended**

B. palmar contact using an arch support under the hand with the fingers extended

C. weight bearing on a closed first

D. total palmer contact with the fingers partially flexed

175. Which surgical technique is commonly performed on a shoulder to repair a defect in the glenoid labrum?

A. acromioplasty

B. **Bankart**

C. inferior capsular shift

D. Magnuson

176. Spasticity of the rhomboids and levator scapulae muscles can facilitate a subluxation of the glenohumeral joint in CVA patient by_______?

A. depressing and rotating the scapula downward

B. **depressing and rotating the scapula upward**

C. elevating and rotating the scapula downward

D. elevating and rotating the scapula upward

177. A C6-C7 quadriplegic is exercising on a mat. The physical therapist assists the patient into a prone on elbows position. The therapist instructs the patient to push their elbows down, tuck their chin and lift their trunk off of the mat while rounding out their shoulders.

This exercise is helpful in strengthening the______?

A. serratus anterior and scapular muscles
B. serratus anterior
C. latissimus dorsi and scapular muscles
D. sacrospinalis and semispibalis

178. A therapist performs a range of motion screening on a 14 years old female. The therapist instructs the patient to stand on their tiptoes. The therapist uses this command to evaluate________?

A. **dorsiflexion and toe extension**
B. doresiflexion and toe flexion
C. plantarflexion and toe extension
D. plantarflexion and toe flexion

179. A physical therapist completes a study which examines the effect of goniometer size on the reliability of passive shoulder of passive shoulder joint measurements. The therapist concludes that goniometric passive measurements of shoulder range of motion can be highly reliable when taken by a single therapist regardless of the size of the goniometer. This study demonstrates the use of _______?

A. interrater reliability
B. intraater reliability
C. **interrater validity**
D. intrarater validity

180. A therapist observes the gait of a patient statues post pneumonia. The therapist notices that the patient walks on his heel. Possible causes of this deviation include all of the following except ______?

A. weakness of the gastrocnemius muscle
B. **tightness of the dorsiflexor muscles**
C. decreased strength in the dorsiflexors
D. pes calcaneus deformity

181. A therapist performs a muscle screening test on a 43 year old female diagnosed with carpal tunnel syndrome. The therapist attemptes to pull the patient's fingers from a position of adduction to abduction. This action is used to test the ______?

A. finger extensors
B. opponens pollicis
C. **dorsal interossei**
D. palmer interossei

182. A developmental evaluation is performed on a normal infant. The infant is able to play on extended arms and attempts reaching for objects. The infant can maintain good alignment of the head and trunk when pulled to a sitting position. The infant will also attempt random stepping when placed in supportive standing. The therapist would expect the infant's age to be ______?

A. 3 month
B. 5 month
C. 7 month
D. **9 month**

183. A physical therapist's primary emphasis when treating a child with________ is aggressive bronchial drainage ,chest vibration and percussion ?

A. cystic fibrosis .
B. **chronic asthma** .
C. respiratory muscle weakness .
D. pneumothorax .

184. A physical therapist observes the standing posture of a patient from a lateral view . If the patient had normal anatomical alignment you would expect the plumb line to fall ___?

A. Posterior to the lobe of the ear .
B. **Anterior to the greater trochanter of the femur** .
C. Slightly anterior to a midline through the knee .
D. Slightly posterior to the lateral malleolus .

185. A patient demonstrates unilateral neglect to the left . The physical therapist's short term goals for the patient include reducing the patient's neglect . Which treatment technique would not be effective in improving the patient's condition ?

A. Have the patient participate in bilateral tasks to increase their body awareness .

B. Place the patient's food on the right side of the tray so the patient can eat by himself .

C. Have the patient observe himself massaging his left upper extremity .

D. Have the patient use a mirror while dressing .

186. A physical therapist instructs a 67 year old female four weeks status post total hip replacement in a home exercise program . Which exercise description would be inappropriate for the patient ?

A. Lie on your involved side and lift the involved leg towards the ceiling .

B. Place a rolled up towel under your knee . Rest the weight of the thigh on the towel roll . Lift heel off the table so that you straighten the knee .

C. Lie on your back with the opposite knee bent . Tighten the thigh muscle, keeping the leg straight and lift leg toward the ceiling .

D. Lie on your back with knees bent . Raise your buttocks of the table

187. A physical therapist performs a vertebral artery test on a patient prior to using traction and mobilization techniques in the upper cervical spine . During the test the patient begins to demonstrate nystagmus, slurring of speech and blurred vision . The therapist should ___?

A. Repeat the vertebral artery test to confirm the original results .

B. **Proceed cautiously with traction and mobilization techniques in the upper cervical spine** .

C. Do not use traction or mobilization techniques in the upper cervical spine .

D. Discharge the patient from physical therapy and refer the patient to their primary physician .

188. A physical therapist straps the ankle of a young athlete with a history of recurrent inversion ankle sprains. The therapist elects to use the closed basket weave strapping technique?

A. One and one half inch adhesive tape is traditionally used .

B. Stirrups should be applied pulling the foot into inversion .

C. The foot should remain in dorsiflexion during strapping .

D. To provide additional support use figure – eights with or without heel locks

189.a 16 year old basketball player is anxious to return to athletic competition following rehabilitation from an Achilles tendon rupture . The athlete is objectively ready to return to competition however he continues to demonstrate a severe preoccupation with reinjury.

The most appropriate response is to ____?

A. Allow the patient to return to basketball without restriction .

B. Inform the patient that he should not participate in basketball this season .

C. Design a functional progression for basketball which allows the patient to progress to higher level activities in a gradual fashion .

D. Discuss other less demanding athletic activities with the patient .

190. A therapist observes random changes in the intensity of an ultrasound generator during patient treatment. The most appropriate response is to ____?

A. Ignore the changes in the intensity since they are normal when using ultrasound .

B. Continue to use the machine at intensity levels less than 1.0 w/cm2.

C. Closely monitor the machine during use.

D. Discontinue use of the machine and contact a service technician.

191. a physical therapist treats a patient using high voltage galvanic stimulation . During treatment the therapist begins to see smoke rising form the machine and smells an unusual odor. The most appropriate immediate, response is to ___?

A. Unplug the machine and label it " Defective – Do Not Use "

B. File an incident report.

C. Contact the appropriate service group.

D. Request an investigation by the biomedical instrumentation department.

192. Palpation can provide a therapist with valuable information during a physical examination. Which of the following statements dose not accurately describe proper palpation technique?

A. Palpation should be performed in a systematic fashion to ensure that all structures are examined .

B. The uninvolved side should be palpated first .

C. **Apply deep pressure initially and progress toward light pressure**.

D. Any differences or abnormalities should be noted .

193. a physical therapy department attempts to develop guidelines for electrical equipment care and service . Which of the following guidelines does not meet acceptable equipment care and service standards ?

A. AC power receptacles and plugs should be hospital grade quality .

B. Electrical equipment should be inspected every 24-36 months .

C. Establish a file of clinical and technical information for each piece of equipment.

D. Appropriate documentation of all inspection and repair activities should be kept for each device.

194. A patient with latissimus dorsi and lower trapezius weakness would have the most difficulty performing which of the following activities ?

A. Four point gait with Loftstrand crutches.

B. Three point gait with a straight cane.

C. Swing through gait with axillary crutches.

D. Wheelchair propulsion.

195. A patient performs bridging in supine as part of his exercise program. The therapist increases the difficulty by applying resistance as the patient maintains the bridge position Resisted bridging reinforces the coactivation of the _____?

A. Hip extensors, abductors, knee flexors.

B. Hip flexors, abductors, knee extensors.

C. **Hip extensors, adductors, knee extensors**.

D. Hip flexors, adductors, knee flexors.

196. a 78 year old female two months status post right total hip replacement is instructed in walking with a cane . The therapist attempts to teach the patient to use the cane in the left hand; however the patient is decidedly right handed and is unable to properly maneuver the cane. The most appropriate response is to ____?

A. Continue to use the cane in the left hand despite the patient's difficulty.

B. Select another appropriate assistive device and continue to instruct the patient in use of the cane on their left side.

C. Instruct the patient to use the cane on their right side permanently.

D. Have the patient use two axillary crutches.

198. After sitting at a computer station for 2-3 hours, an individual reports experiencing a sharp, localized pain in the left arm. When asked to show the location of the pain, the individual points to the area of the insertion of the deltoid. The pain disappears when the individual stands up and walks around briefly. Which of the following interventions is MOST likely to correct the problem**?**

a. Isometric strengthening of the deltoid
b. Lumbar extension exercises in prone
c. Instruction in correct postural alignment in sitting
d. Instruction in shoulder active range of motion exercises

199. A patient with a medullary level vascular lesion has increased vagal nerve activity. Which of the following descriptions BEST represents the cardiovascular effects that occur when the patient transitions from supine to standing**?**

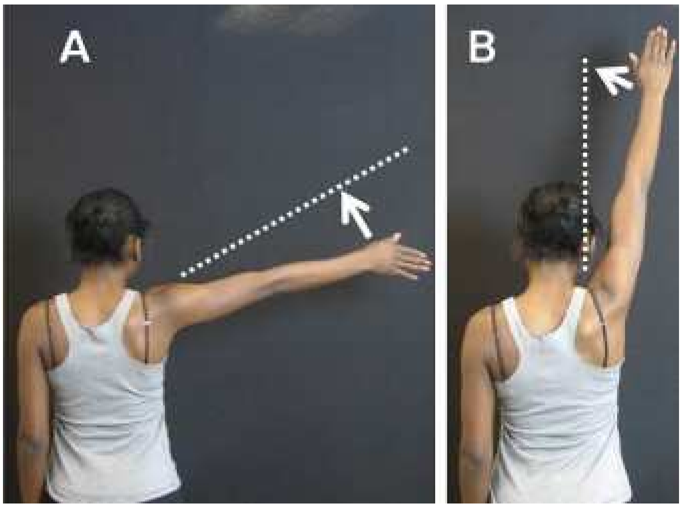
a. Rise in blood pressure and no change in heart rate
b. Drop in blood pressure and no change in heart rate
c. Rise in blood pressure and an increase in heart rate
d. Drop in blood pressure and an increase in heart rate

200. A patient has used crutches with a partial weight-bearing toe touch gait for the past 3 months. When progressed to full weight bearing, the patient is unable to demonstrate a heel-toe gait sequence with the involved extremity. Which of the following disorders is the MOST likely origin of the gait abnormality?

a. Plantar fasciitis
b. Fibular (peroneal) nerve palsy
c. Heel cord tightness
d. Hammer toe

201. While abducting the shoulder, the patient in the photograph denies pain while moving the arm through the range indicated in photograph A, but reports increasing pain severity as the arm moves into the range indicated by photograph B. Which of the following disorders is MOST likely present?

a. Acromioclavicular joint lesion
b. Subacromial bursitis
c. Infraspinatus tendinopathy
d. Partial tear of the supraspinatus



202. During a physical therapy evaluation, a patient with a sprain of the deltoid ligament of the ankle reported pain with palpation of the affected area and with ankle motion that stresses the ligament. To determine any change in the patient's pain level during subsequent treatments, a physical therapist assistant should perform which of the following actions?

a. Palpate anterior to the lateral malleolus, and passively plantar flex the ankle.
b. Palpate inferior to the medial malleolus, and passively evert the ankle.
c. Palpate over the sinus tarsi, and passively invert the ankle.
d. Palpate deep to the Achilles tendon, and passively dorsiflex the ankle.

203. Which of the following instructions would be MOST appropriate to give a patient who is learning pursed-lip breathing?

a. Exhale through pursed lips while contracting abdominal muscles.
b. Exhale by blowing air out forcefully between pursed lips.
c. Exhale by relaxing air out through pursed lips.
d. Exhale in quick short puffs through pursed lips.

204. A patient with impaired kinesthesia who is performing active range of motion exercises of the upper extremities will MOST likely require which of the following supplemental interventions?

a. Visual cues while performing the exercises
b. Pain-relieving modalities prior to performing the exercises
c. Trunk stabilization while performing the exercises
d. Stretching prior to performing the exercises

205. If angle 1 in the photograph is greater than 20º, which of the following complications is the patient most likely to develop?

a. Patellofemoral tracking disorder
b. Genu recurvatum
c. Lateral collateral ligament sprain
d. Medial meniscal lesion

206. A physical therapist is working in an outpatient orthopedic clinic. During the patient's history the patient reports, "I tore 3 of my 4 Rotator cuff muscles in the past." Which of the following muscles cannot be considered as possibly being torn?

A. Teres minor
B. **Teres major**
C. Supraspinatus
D. Infraspinatus

207. A physical therapist working on an ICU unit, notices a patient is experiencing SOB, calf pain, and warmth over the posterior calf. All of these may indicate which of the following medical conditions?

A. **Patient may have a DVT.**
B. Patient may be exhibiting signs of dermatitis.
C. Patient may be in the late phases of CHF.
D. Patient may be experiencing anxiety after surgery.

208. A physical therapist is caring for a patient who has recently been diagnosed with fibromyalgia and COPD. Which of the following tasks should the physical therapist delegate to an aide?

A. **Transferring the patient during the third visit.**
B. Ambulating the patient for the first time.
C. Taking the patient's vital sign while setting up an exercise program.
D. Educating the patient on monitoring fatigue.

209. A physical therapist is instructing a person who had a left CVA and right lower extremity hemiparesis to use a quad cane. Which of the following is the most appropriate gait sequence?

A. **Place the cane in the patient's left upper extremity, encourage cane, then right lower extremity, then left upper extremity gait sequence.**
B. Place the cane in the patient's left upper extremity, encourage cane, then left lower extremity, then right upper extremity gait sequence.
C. Place the cane in the patient's right upper extremity, encourage cane, then right lower extremity, then left upper extremity gait sequence.
D. Place the cane in the patient's right upper extremity, encourage cane, then left lower extremity, then right upper extremity gait sequence.

210. Electrical stimulating currents may produce the following effects:

e. muscle contraction
f. net ion movement
g. decrease in pain
h. all of the above

211. A physical therapist is caring for a patient in the step down unit. The patient has signs of increased intracranial pressure. Which of the following is not a sign of increased intracranial pressure?

A. Bradycardia
B. **Increased pupil size bilaterally**
C. Change in LOC

D. Vomiting

212. The charge nurse on a cardiac unit tells you a patient is exhibiting signs of right-sided heart failure. Which of the following would not indicate right-sided heart failure?

A. Nausea
B. Anorexia
C. Rapid weight gain
D. **SOB (shortness of breath)**

213. A patient has recently been diagnosed with symptomatic bradycardia. Which of the following medications is the most recognized for treatment of symptomatic bradycardia?

A. Questran
B. Digitalis
C. Nitroglycerin
D. **Atropine**

214. Which of the following arterial blood gas values indicates a patient may be experiencing a condition of metabolic acidosis?

A. PaO2 (91%)
B. **Bicarbonate 159**
C. CO(2) 48 mm Hg
D. pH 7.33

215. A patient has suffered a left CVA and has developed severe hemiparesis resulting in a loss of mobility. The physical therapist notices on assessment that an area over the patient's left elbow appears as non-blanchable erythema and the skin is intact. The physical therapist should score the patient as having which of the following?

A. **Stage I pressure ulcer**
B. Stage II pressure ulcer
C. Stage III pressure ulcer
D. Stage IV pressure ulcer

216. A physical therapist is performing an evaluation on a patient that has fallen off a fork lift and injured his right arm. The patient was evaluated by an MD 3 days ago. The patient reports, "Yesterday my arm started to go numb and turned purple." A poor arterial pulse is found in the arm upon evaluation. Which step should the physical therapist take first?

A: Notify the MD of the changes immediately
B: Begin light exercise program per MD orders
C: Apply E-stim and Ice to the extremity
D: Special test the right arm

217. A physical therapist is reviewing a patient's medication during an evaluation. Which of the following medication would be contraindicated if the patient were pregnant?

A: Coumadin
B: Celebrex
C: Catapress
D: Habitrol

218. A physical therapist is reviewing a patient's PMH. The history indicates photosensitive reactions to medications. Which of the following drugs has not been associated with photosensitive reactions?

A: Nitrodur
B: Sulfonamide
C: Noroxin
D: Bactrim

219. A patient has a diabetic ulcer on his right foot first ray region. The ulcer is in a chronic state of tissue inflammation and the patient is extremely obese. The wound is not infected at this time; however the patient is unable to ambulate at this time without applying pressure to the first ray region. Which of the following interventions would be the most beneficial to the patient?

A: Whirlpool 10 minutes followed by sharps debridement
B: Have the patient fitted to offload pressures on the first ray in a week.
C**: Review the role of glucose and tissue healing with the patient.**
D: Have the patient perform NWB gait with Axillary crutches with the extremity.

220. A thirty five year old male has been an insulin-dependent diabetic for five years and now is unable to urinate. Which of the following would you most likely suspect?

A: Atherosclerosis
B**: Diabetic nephropathy**
C: Autonomic neuropathy
D: Somatic neuropathy

221. You are taking the history of a 14 year old girl who has a (BMI) of 18. The girl reports inability to eat, induced vomiting and severe constipation. Which of the following would you most likely suspect?

A: Multiple sclerosis
B**: Anorexia nervosa**
C: Bulimia
D: Systemic sclerosis

222. A patient has a Right T8 facet joint that has become extremely tight. Which of the following movements with stretch the joint with the best technique?

A: L Trunk Rotation and Extension
B**: L Trunk Rotation and Flexion**
C: R Trunk Rotation and Extension
D: R Trunk Rotation and Flexion

223. A fifty-year-old blind and deaf patient has been assigned to you for evaluation. As the lead physical therapist your primary responsibility for this patient is?

A**:** Let others know about the patient's deficits
B**:** Communicate with your supervisor your concerns about the patient's deficits.
C**:** Continuously update the patient on the social environment.
D**: Provide a secure environment for the patient**.

224. Your patient is getting discharged from a SNF facility. The patient has a history of severe COPD and PVD. The patient is primarily concerned about their ability to breath easily. Which of the following would be the best instruction for this patient?

A: Deep breathing techniques to increase O2 levels.
B: Cough regularly and deeply to clear airway passages.
C: Cough following bronchodilator utilization
D: Decrease CO2 levels by increase oxygen take output during meals.

225. Which of the following is the best exercise to correct a Trendelenburg gait pattern?

A: Bridging
B: Bridging with Resisted Abduction
C: Bridging with Straight Leg Raise
D: Squats

226. A physical therapist is caring for an infant that has recently been diagnosed with a congenital heart defect. Which of the following clinical signs would most likely be present?

A: Slow pulse rate
B: Weight gain
C: Decreased systolic pressure
D: Irregular WBC lab values

227. A mother has recently been informed that her child has Down's syndrome. You will be evaluating the patient in the afternoon. Which of the following characteristics is not associated with Down's syndrome?

A: Simian crease
B: Brachycephaly
C: Oily skin
D: Hypotonicity

228. A patient asks a physical therapist, "My doctor recommended I increase my intake of folic acid. What type of foods contain folic acids?"

A: Green vegetables and liver
B: Yellow vegetables and red meat
C: Carrots
D: Milk

229. A patient has developed trochanteric bursitis that has gone untreated for 5 weeks. The patient is 34 years old. Which of the following Ultrasound settings is most appropriate?

A: Pulsed US at 3MHz
B: Pulsed US at 1MHz
C: Continuous US at 3MHz
D: Continuous US at 1MHz

230. A child is 5 years old and has been recently admitted into the hospital. According to Erickson which of the following stages is the child in?

A: Trust vs. mistrust
B: Initiative vs. guilt
C: Autonomy vs. shame
D: Intimacy vs. isolation

231. When you are taking a patient's history, she tells you she has been depressed and is dealing with an anxiety disorder. Which of the following medications would the patient most likely be taking?

A: Elavil
B: Calcitonin
C: Pergolide
D: Verapamil

232. A patient's chart indicates a history of hyperkalemia. Which of the following would you not expect to see with this patient if this condition were acute?

A: Decreased HR
B: Paresthesias
C: Muscle weakness of the extremities
D: Migranes

233. A patient's chart indicates a history of meningitis. Which of the following would you not expect to see with this patient if this condition were acute?

A: Increased appetite
B: Vomiting
C: Fever
D: Poor tolerance of light

234. A fragile 87 year-old female has recently been admitted to the hospital with increased confusion and falls over last 2 weeks. She is also noted to have a mild left hemiparesis. Which of the following tests is most likely to be performed?

A: FBC (full blood count)
B: ECG (electrocardiogram)
C: Thyroid function tests
D: CT scan

235. A physical therapist is evaluating an adult that has recently been diagnosed with hypokalemia. Which of the following clinical signs would most likely not be present?

A: Leg cramps
B: Respiratory distress
C: Confusion
D: Flaccid paralysis

236. A physical therapist is evaluating an adult that has recently been diagnosed with respiratory acidosis. Which of the following clinical signs would most likely not be present?

A: CO2 Retention
B: Dyspnea
C: Headaches
D: Tachypnea

237. A physical therapist is caring for an adult that has recently been diagnosed with respiratory alkalosis. Which of the following clinical signs would most likely not be present?

A: Anxiety attacks
B: Dizziness
C: Hyperventilation cyanosis
D: Blurred vision

238. A physical therapist is evaluating a patient that has had damage to the ulnar nerve. Which of the following muscles would most likely show signs of weakness?

A: Soleus
B: Triceps
C: Brachioradialis
D: Adductor Pollicus

239. A patient has been on long-term management for CHF. Which of the following drugs is considered a loop dieuretic that could be used to treat CHF symptoms?

A: Ciprofloxacin
B: Lepirudin
C: Naproxen
D: Bumex

240. A patient has recently been diagnosed with polio and has questions about the diagnosis. Which of the following systems is most affected by polio?

A: PNS
B: CNS
C: Urinary system
D: Cardiac system

241. A physical therapist is educating a patient about right-sided heart deficits. Which of the following clinical signs is not associated with right-sided heart deficits?

A: Orthopnea
B: Dependent edema
C: Ascites
D: Nocturia

242. A physical therapist is reviewing a patient's medication. The patient is taking Digoxin. Which of the following is not an effect of Digoxin?

A: Depressed HR
B: Increased CO
C: **Increased venous pressure**
D: Increased contractility of cardiac muscle

243. A patient's chart indicates the patient is suffering from Digoxin toxicity. Which of the following clinical signs is not associated with digoxin toxicity?

A: Ventricular bigeminy
B: Anorexia
C: Normal ventricular rhythm
D: Nausea

243. Which of the following is the key risk factor for development of Parkinson's disease dementia?

A: History of strokes
B: Acute headaches history
C: Edward's syndrome
D: Use of phenothiazines

244. Skeletal system is the biological system providing support in living organisms. It is a rigid framework that provides protection and structure in many types of animals as well as living beings, particularly those of the phylum Chordata and of the super phylum Ecdysozooa .It changes with the approximate varying age. Which is the true statement among the following?

a. Increase in bone mass and density.

b. Cartilage become strong and strong and defragment occurs.

c. Bone marrow increases with red blood cell production.

d**. Ligaments and bones become weak and susceptible**.

245. Progressive Resistance Strength Training results in improvements to muscle strength and some aspects of functional limitation, such as gait speed, in older adults. However, based on current data, the effect of PRT on physical disability remains unclear. Further, due to the poor reporting of adverse events in trials, it is difficult to evaluate the risks associated with PRT. Which is the false statement among the following?

a. Significant improvements noted in frail, institutional sized 80 and 90 years.

b. Improvements in strength correlate to improved functional abilities.

c. **Cervical flexor muscles exercise is not effective in reducing myoelectrincreases manifestations of superficial cervical flexor muscle fatigue in elders.**

d. Increase in strength noted in older adults with isometric and progressive resistive exercise regimes.

246. Brain Morphology contribute to the significant association between the incidence of schizophrenia, increased left or mixed handedness, reduction in cerebral asymmetry and anomalies .Studies have shown that prenatal stress in elders can mimic these developments and behavioral alterations .They shows a reduced propensity to social interaction, increased anxiety or novel situation .Which is the false change among the following?

a**.** **Behavioral Abnormalities occurs due to Intervertebral discs: flatten, less resilient due to loss of water content (30% loss by age 65) and loss of collagen elasticity; trunk length**.

b. Generalized cell loss in cerebral cortex: especially frontal and temporal lobes, association areas.

c. Presence of lipofuscins, senile or neuritic plaques, and neurofibrillary tangles (NFf).

d. Decreased cerebral blood flow and energy metabolism.

247. The spinal cord is a long, thin, tubular bundle of nervous tissue and support cells that extends from the brain (the medulla specifically). The brain and spinal cord together make up the central nervous system. The spinal cord extends down to the space between the first and second lumbar vertebrae; it does not extend the entire length of the vertebral column. With ageing which is the approximate change seen in older people?

a. Slowed nerve conduction velocity: sensory greater than motor.
b. The disks do not lose their capacity to cushion.
c. Loss of sympathetic fibers: may account for diminished, autonomic stability.
d. Loss of motoneurons results in increase in size of remaining motor units.

248. Tremor is caused by abnormalities in areas of the brain that control movement and does not occur as the result of disease. It is involuntary trembling involving a certain part of the body. Essential tremor (ET) is tremor that occurs with purposeful movement. Incidence is highest in people over the age of 60.Which is a false statement about tremor found in the elder people above the age of 60?

a. Benign, slowly progressive; in late stages may limit function

b. Isolated symptom, particularly in hands, head and voice

c. Characterized as postural or kinetic, rarely resting

d**. Do not get worse with stress, caffeine.**

249. Treatment may not be necessary unless the tremors interfere with your daily activities or cause embarrassment. Treatment depends on the cause. Tremors caused by a medical condition such as hyperthyroidism will likely get better when the condition is treated. Which is the incorrect statement to treat tremors?

a. Improve health: diet, smoking cessation.

b. Correction of medical problems: improve cerebral blood flow.

c. Avoid Propranolol and Primidone.

d. Allow for increased cautionary behaviors: provide adequate explanation, demonstration when teaching new movement skills.

250. Essential tremor is characterized by shaking during movement; the symptoms of Parkinson's disease include involuntary shaking at rest, muscle stiffness and freezing. Parkinson's disease is caused by a deficiency of the brain chemical dopamine, which is necessary for smooth and controlled muscular movement. Which is the not an appropriate movement of elder in tremor?

a. Both reaction time and movement time are increased.
b. Speed and coordination are decreased; increased difficulties with fine motor control.
c. **Gait abnormalities and Excellent mobility**.
d. Complicated movements require more preparation, longer reaction and movement times.

251. The brain and nervous system receive input from body parts as well as from the outside world. The central nervous system is also a means of transmitting messages throughout the body and functions somewhat like a computer system. The messages that are transmitted, however, affect functions such as muscle movement, coordination, learning, memory, emotion, behavior and thought. Which is not an age related change in the in elder's sensory systems?

a. **Person needs more light to see.**

b. Strain social interactions.

c. Leads to decreased functional mobility, risk of injury.

d. Leads to sensory deprivation, isolation, disorientation, confusion, appearance of senility.

252. Significant losses to eyesight can reduce quality of life and threaten ability to live independently at home and in the community. There is a general decline in visual activity; gradual prior to sixth decade, rapid decline between ages 60 and 90.Which of the following is not a symptom of vision loss?

a. Decreased pupillary responses, size of resting pupil increases.
b. Decreased ability to adapt to dark and light.
c. Decreased sensitivity of corneal reflex: less sensitive to eye injury or infection.
d. **An attached retina**.

252. It is important for persons to give care to older adults to understand how age affects the eye, recognize the most common eye problems, and learn ways to deal with poor eyesight. Which is one of the false ways to come over this?

a. **Avoid the usage of Vitamin A.**

b. Work in adequate light, reduce glare; avoid abrupt changes in light.

c. Assess for use of glasses, need for environmental adaptations.

d. Use warm colors (yellow, orange, red) for identification and color coding.

253. Pathology is the study and diagnosis of disease through examination of organs, tissues, bodily fluids, and whole bodies. Pathology also encompasses the related scientific study of disease processes, called general pathology. Which is not a reason in the following that causes loss of vision due to pathology?

a. Senile macular degeneration
b. Parkinson 's disease.
c. Menieres Disease.
d. Onchocerciasis.

254. The ability to hear clearly declines with age. Hearing loss often begins at a young age and progresses slowly during the 20s, 30s, and 40s. Most people do not notice hearing loss until they are in their 50s or 60s, when they begin to have a hard time hearing high frequency sounds. Which of the following is not a symptom of hearing problem?

a. Significant changes in sound sensitivity, understanding of speech.
b. Minimal degenerative changes of bony joints.

c. Buildup of cerumen (ear wax) may result in conductive hearing loss

d. **Oculomotor responses diminished.**

255. Family caregiver service providers and health care professionals often feel frustrated when trying to communicate with an older person who cannot hear well. Listening and following a conversation is tiring for someone with poor hearing. Which is not a hearing loss from the following?

a. **Arkinson's disease or Alzheimer's disease.**

b. Sensorineural.

c. Presbycusis.

d. Conductive.

255. Inappropriate actions based upon missed information can lead to a feeling of paranoia. When older persons fail to answer when spoken too or if they give inappropriate answers to questions, they are sometimes considered confused or demented. Significant loss of hearing can cause older adults to feel cut off from friends and family. Which is one of the false ways to come over the hearing problem?

a. Measure air and bone conduction: Rinne test, Weber test.
b. Determine use of hearing aids; check for proper functioning.
c. Avoid Maskers.

d. Orient person to topics of conversation they cannot hear to reduce paranoia, isolation.

256. Pathology is the study and diagnosis of disease through examination of organs, tissues, bodily fluids, and whole bodies. Pathology also encompasses the related scientific study of disease processes, called general pathology. Which is not a reason in the following that causes loss of vision due to pathology?

a. Hypothyroidism

b. Paget's disease.

c. Otosclerosis

d. **Glaucoma.**

257. Vestibular balance control shows degenerative changes in otoconia of utricle and saccule; loss of vestibular hair-cell receptors; decreased number of vestibular neurons. Which is not a factor of age change in vestibular balance control?

a. Reduced function of vestibular ocular reflex (VOR).

b. Altered sensory organization: older adults more dependent upon somatosensory inputs for balance.

c. Postural response patterns for balance are disorganized.

d. **Increased thresholds in vibratory sensibility.**

258. The human balance system depends on the inner ear, the eyes, and the muscles and joints to transmit reliable information about the body's movement and orientation in space. Which is the disease not caused by additional loss of vestibular sensitivity with pathology?

a. Cerebrovascular disease.

b. Cerebellar dysfunction.

c. **Presbycusis.(hearing loss)**

d. Benign paroxysmal positional vertigo.

258. The somatosensory system is a diverse sensory system comprising the receptors and processing centers to produce the sensory modalities such as touch, temperature, proprioception(body position), and nociception (pain). The sensory receptors cover the skin and epithelia, skeletal muscles, bones and joints, internal organs, and the cardiovascular system. Which is not a leading aging change due to Somatosensory aging change?

a. Proprioceptive losses, increased thresholds in vibratory sensibility.
b. Cutaneous pain thresholds increased: greater changes in upper body areas.
c. Loss of joint receptor sensitivity; losses in lower extremities, cervical joints may contribute to loss of balance.
d. **Loss of power**.

259. The somatosensory system consists of ascending pathways from the body to the postcentral gyrus in the cerebral cortex, namely the Dorsal Column Medial Lemniscal pathway, the Ventral Spinothalamic pathway, ventral and dorsal spinocerebellar tracts. Areas of the part of the human brain map to certain areas of the body, dependent on the amount or importance of somatosensory input from that area. Which is not a disease caused by disorder of Somatosensory system?

a. **Otosclerosis.**
b. Diabetes, peripheral neuropathy
c. CVA, central sensory losses
d. Peripheral vascular disease, peripheral ischemia.

260. As mentioned above, Areas of the part of the human brain map to certain areas of the body, dependent on the amount or importance of somatosensory input from that area. Which is not a compensatory strategy to recover somatosensory system?

a. Use touch to communicate: maximize physical contact.
b. Allow extra time for responses with increased thresholds.
c. **Measure air and bone conduction.**
d. Teach compensatory strategies to prevent injury to anesthetic limbs, falls..

261. As the disorders of smell and taste are rarely life threatening, they may not receive close medical attention. Yet these disorders can be frustrating because they can affect the ability to enjoy food and drink and to appreciate pleasant aromas. Which one of the following can be a reason for taste and smell disorder?

a. **Head injury**

b. Leg injury.

c. Diabetes.

d. Headache.

262. Although our sense of smell is our most primal, it is also very complex. For humans, odors communicate a variety of messages, depending on the odor and the person receiving it. Which is not a clinical implications or a compensatory strategy to prevent smell and taste disorder?

a. Older adults frequently increase use of taste enhancers

b. Decreased home safety.

c. Examine ability to identify odors, tastes.

d. **Increase food to enjoy food and taste**.

263. Most common mental disorders affect cognitive functions, mainly memory processing, perception and problem solving. The most direct cognitive disorders are amnesia, dementia and delirium. Cognitive disorders are mental disorders that develop on the basis of cognitive mental disorder perspective. Which is not an age related change in cognition among the following?

a. decline in measures of intelligence occur in the year immediately preceding death

b. (by age 39); require longer times to complete tasks.

c. Impairments of memory are typically noted in short-term memory; long-term memory retained.

d**. Changes in intellectual abilities typically show up till mid 60s.**

264. Learning is acquiring new knowledge, behaviors, skills, values or preferences. It may involve processing different types of information. Learning functions can be performed by different brain learning processes, which depend on the mental capacities. Which of the following factor is an obstacle in learning for the older people?

a. **Dentures.**

b. Increased cautiousness.

c. Anxiety.

d. Interference from prior learning. .

265. Lifelong learning is the process of keeping your mind and body engaged, at any age, by actively pursuing knowledge and experience. The pursuit of knowledge through lifelong learning has wonderful benefits for adults 50-plus. Which of the following factor is not responsible to intervene to slow changes or enhance learning in older age?

a. Increase mental activity.

b. Improve health

c. Reduction of stress: counseling and family support.

d. **Isolate from crowd.**

266. The cardiovascular system is a closed tubular system in which the blood is propelled by the heart. The system has two circuits, the pulmonary circuit and the systemic circuit. Each circuit has arterial, capillary, and venous components. t is essentially a long, closed tube through which blood moves in a double circuit — one through the lungs (pulmonary circulation) and one through the rest of the body (systemic circulation).Which one of the following is an age related to cardiovascular disorder?

a. Cardiac valves thicken and stiffen.

b. Degeneration of heart muscle with accumulation of lipofuscins.

c. **Altered pulmonary gas exchange.**

d. Decline in neurohumoral control.

267. The cardiovascular system is formed by the heart, arteries, and veins. In connection to the respiratory system, the cardiovascular system provides oxygen to cell and collect carbon dioxide (CO_2). It also helps on the transportation of hormones, wastes. It is responsible for the circulation of the blood that carries all the substances to and from the cell. With the blood circulating the temperature of the body is also affected. Which of the following is not a correct clinical implication for cardio logical system?

a. Blunted, decrease in heart rate acceleration, decrease maximal oxygen uptake and heart rate.

b**. Respiratory responses to exercise similar to younger adult at low and moderate intensities; at higher intensities.**

c. Decreased stroke volume due to decreased myocardial contractility.

d. Orthostatic hypotension.

268. The Integrated system of organs involved in the intake and exchange of oxygen and carbon dioxide between an organism and the environment. In humans, the diaphragm and, to a lesser extent, the muscles between the ribs generate a pumping action, moving air in and out of the lungs through a system of pipes (conducting airways), divided into upper and lower airway systems. Which one of the following is not a pulmonary system age related disorder?

a. Loss of lung elastic recoil, decreased lung.

b**. Changes in blood vessels**

c. Forced expiratory volume (air flow) decrease

d. Blunted defense/immune responses.

269. In the following which is not a clinical implication of Pulmonary System Disorder?

a. Cough mechanism is impaired.

b. **Gag reflex is decreased, increased risk of aspiration.**

c. Provide augmented feedback through appropriate sensory channels.

d. Recovery from respiratory illness.

270. Complete cardiopulmonary examination prior to commencing an exercise program is essential in older adults due the high incidence of cardiopulmonary pathologies. Which is the incorrect mode for the Interventions to slow or reverse changes in cardiopulmonary system?

a. Selection of appropriate exercise tolerance testing protocol (ETT) is important.

b. Individualized exercise prescription essential.

c. Aerobic training programs can significantly improve cardiopulmonary function in the elderly.

d. **Ignore circuit training**.

271. Aerobic training programs can significantly improve cardiopulmonary function in the elderly age. Which of the following is not an advantage of aerobic training?

a. Improves recovery heart rates.

b. Increase maximum ventilatory capacity: vital capacity.

c. Reduces breathlessness, lowers perceived exertion.

d. **Increase systolic blood pressure.**

272. The integumentary system has a variety of functions; it may serve to waterproof, cushion and protect the deeper tissues, excrete wastes, regulate temperature and is the attachment site for sensory receptors to detect pain, sensation, pressure and temperature. In humans the integumentary system additionally provides vitamin D synthesis. what cannot be found as a major skin change in integumentary system?

a. Dermis thins with loss of elastin.

b. Decreased sebaceous activity and decline in hydration.

c. **Nails grow fast.**

d. skin appears dry and wrinkled.

273. What is the role of a primary care physician (PCP) in Primary Care .Primary Care is the Basic or "first level", health care provided on an outpatient basis.

a **Gate keeper**

b. Physical therapists

c. Nurse

d. physician specialists

274. Chronic Care Facility (long term care facility) is the long term care facility providing services to patients for 60 days or longer. What type of patient requires Chronic Care Facility?

a. Patients that are not in an acute phase of illness, but require skilled care on an inpatient basis.

b**. patients with permanent or residual disability caused by a non-reversible pathological health condition**

c. provided to out patients

d. patients with short term illness or health problem.

275. Custodial Care Facility means patient care that is not medically required but necessary for the patient who is unable to care for him/herself. Custodial care is provided by which of the following?

a. **non medical support staff**

b. Providers may be physicians, physician assistants, nurse practitioners, physical therapists, or others

c. Provided by medical and nursing services as well as rehabilitative services

d. Provided by highly specialized physicians in hospital setting.

276. Home Health Care is provided to individuals and their families in their homes. It is provided by a Home Health Agency, which may be governmental; voluntary or private; non-profit or for-profit. This is the reimbursable equipment in home health care;

a. **wheelchairs, commodes, hospital beds**

b. grab bars

c. long handled utensils

d. lights

277. Ultrasound is produced by the conversion of mechanical energy produced by sound waves at frequencies between 85 KHz and 3 MHz and delivered at intensities between 0 and 3 W/cm2 is absorbed by body tissues and changed to thermal energy. What converts electrical energy into acoustical energy for ultrasound production?

a. Alternating voltage

b. **Transducer**

c. Crystal

d. Oscillating sound wave

278. Beam non uniformity ratio (BNR). This is the ratio of the spatial peak intensity to the spatial average intensity. What is the ideal Beam non uniformity ratio?

a. 1:2

b. 1:3

c. 1:5

d. **1:1**

289. Overload principle is used to enhance physiologic improvement and bring about a training change. Which is the correct statement regarding overload principle?

a. **The appropriate overload for each person can be achieved by manipulating combinations of training frequency, intensity, and duration**.
b. Specific exercise elicits specific adaptations creating specific training effects.
c. A cardio respiratory training effect can be achieved at a rating of "somewhat hard" or "hard" (13 to 16 on the original Borg scale of 6 to 19).
d. Training level or target heart rate (THR) can be established at 70% of maximum to increase aerobic capacity.

290. FlIT equation includes factors that affect training; frequency, intensity, time and type. Intensity is interrelated with both duration (time) and frequency. In this aspect frequency is the number of exercise sessions per week. Which is the incorrect statement relating to frequency at FIIT?

If the intensity is constant, the benefit from 2 versus 4 or 3 versus 5 times per week is the same.

a. Less than 2 days per week does not produce adequate changes in aerobic capacity or body composition.
b. For weight loss, 5-7 days per week increases the caloric expenditure more than 2 days per week.
c. The Karvonen formula is used to predict heart rate reserve.

291. The intensity and duration of the work intervals and the length of the rest periods indictates the training response. Very short, all-out bouts of work coupled with longer rest periods are used for speed and speed endurance development. Which is the incorrect statement about intensity of cardiovascular endurance?

a. Relative intensity for an individual is calculated as a percentage of the maximum function.
b. The V02 max or HR max can be measured directly or indirectly based on different methods.
c. Rating of Perceived Exertion (RPE) can be used to evaluate training at relative levels.
d. The Karvonen formula is used to predict heart rate reserve.

292. Duration of Exercise is the second component and refers to the time you've spent exercising. The cardio work out plan, not including the warm-up and cool-down, should vary from 20-60 minutes to gain significant cardiorespiratory work out plans and fat burning benefits. Which statement is incorrect in terms of duration of exercise?

a. Multiple sessions of short durations are also indicated when intensity is limited by environmental conditions.
b. Obese individuals should exercise at lower durations and longer intensities.
c. Obesity increases the mechanical work of the heart and can lead to cardiac and left ventricular dysfunction.
d. Duration is increased when intensity is limited.

293. Pulmonary rehabilitation traditionally consists of endurance-type exercises that do not maximally promote strength. This limits the effectiveness of pulmonary rehabilitation because improvement in strength is a key factor in maintaining function and independence, especially in older, frail adults. Keeping this in mind which statement is incorrect in context of Training Strategies to Develop Pulmonary Endurance?

a. Pulmonary endurance is related to the ventilation of the lungs and oxygen consumption.
b. When exercising in humid versus dry environments, the exercise-induced asthmatic response is considerably reduced.
c. The problem is rare in activities that require only short bursts of activity, such as baseball and is more likely to occur in endurance activities such as soccer.
d. In severe pulmonary disease, the cost of breathing can reach 80% of the total exercise oxygen consumption.

294. Exercise induced asthma (EIA) can occur when the normal initial bronchodilatation is followed by bronchoconstriction. The reduction in airflow from airway obstruction affects the ability of the lungs to provide oxygen to exercising muscles. Which statement is incorrect in terms of EIA?

a. When a person mouth breathes, the air is cold and dry, contributing to the bronchoconstriction.
b. The problem is rare in activities that require only short bursts of activity.
c. When exercising in humid versus dry environments,the exercise-induced asthmatic response is considerably reduced.
d. EIA is an acute, reversible airway obstruction that develops 25 to 35 minutes after strenuous exercise.

295. Aerobic training (cardio respiratory endurance training) can result in higher fitness levels for healthy individuals, slow the decrease in functional capacity in the elderly, and recondition those that have been ill or have chronic disease. Which statement is a Positive effect of aerobic training on the cardiovascular and respiratory systems?

a. Increase total hemoglobin and oxygen delivery capacity.
b. Increase cardiac output and stroke volume.
c. Increase resting and sub maximal exercise heart rates.
d. Increase total hemoglobin and oxygen delivery capacity.

296. Continuous training is a sub maximal energy requirement that can be prolonged for 20 to 60 minutes without exhausting the oxygen transport system. Which statement is incorrect in terms of continuous training**?**

a. In healthy individuals, continuous training is the most effective way to improve endurance.
b. Overload can be accomplished by increasing the exercise duration.
c. Work rate is increased progressively as training improvements are achieved.
d. The relief interval can be passive or active; its duration ranges from a few seconds to several minutes.

297. 8 Circuit training is a combination of high-intensity aerobics and resistance training designed to be easy to follow and target fat loss, muscle building and heart fitness. An exercise "circuit" is one completion of all prescribed exercises in the program. When one circuit is complete, one begins the first exercise again for another circuit. Which statement is incorrect in terms of circuit training?

a. Several exercise modes can be utilized involving large and small muscle groups both statically and dynamically.
b. Circuit training improves endurance and strength by stressing the aerobic and anaerobic energy systems.
c. **Provides maximum time for an instructor to ensure that the activity remains safe**.
d. Can be customized for specificity; easy to adapt to your sport.

298. Interval training includes an exercise period followed by a prescribed rest interval. It is perceived to be less demanding than continuous training and tends to improve strength and power more than endurance. Which is the incorrect statement in terms of interval training?

a. **Minimum amount of high-intensity work can be achieved**.
b. The longer the work interval, the more the aerobic system is stressed and the duration of the rest period is not important.
c. The relief interval can be passive or active; its duration ranges from a few seconds to several minutes.
d. In a short work interval, a work-recovery ratio of 1: 1 to 1:5 is appropriate to stress the aerobic system.

299. The purpose of a warm-up and cool down is to encourage the adjustments to occur gradually, by commencing your exercise session at an easy level and increasing the intensity gradually. Each exercise session includes a 5-15 minute warm-up and a SIS minute cool-down period. Which statement is incorrect in terms of Warm up and cool down sessions?

a. The warm-up period prevents the heart and circulatory system from being suddenly taxed.
b. Longer warm-up and cool-down periods may be needed for deconditioned or older individuals.
c. **The cool-down period consists of exercising at a higher intensity**.
d. Primes your nerve-to-muscle pathways to be ready for exercise.

300. Some weight trainers perform light, high-repetition exercises in an attempt to "tone" their muscles without increasing their size Which is not a common error related to Common Errors Associated with Muscular, Cardiovascular and Pulmonary Endurance Training?

a. Lack of exercise tolerance testing (ETT) before the exercise prescription is determined could result in a training program set too high or too low for an individual.
b. Increasing intensity too fast can create a problem for an individual during endurance training.
c. Exercising at too intense a level can use the anaerobic energy system not aerobic system.
d. There is adequate time to prepare for or recover from higher intense activity.

301. The body's adaptation to high altitude exercise helps significantly but doesn't fully compensate for the lack of oxygen. There is a drop in VO2 max of 2% for every 300 m elevation above 1500 m even after allowing for full acclimatization. Which statement is incorrect in terms of exercises at high altitudes?

a. The partial pressure of oxygen is reduced resulting in poor oxygenation of hemoglobin.
b. Body fluids can be rapidly increases through desertion.
c. At altitudes of 6,000 feet (2,000 m) or higher there can be a noticeable drop in performance of aerobic activities.
d. Reduction in CO2 from hyperventilation results in more alkaline body fluids.

302. Altitude training traditionally referred to as altitude camp, is the practice by some endurance athletes of training for several weeks at high altitude, preferably over 2,500 m (8,000 ft) above sea level. Which is an incorrect statement defining Adjustments or acclimatization need to be done for higher altitude exercises?

a. Takes 2 weeks at 2,300m and an additional week for every additional 600m in altitude.
b. **Adjustments fully compensate for altitude**.
c. Changes in local circulation may facilitate oxygen transport.
d. Training at altitude does not provide any improvement in sea-level performance.

303. Exercise in hot weather puts extra stress on your heart and lungs. Both the exercise itself and the air temperature increase your body temperature. Which is a false statement related to exercise in hot weather?

a. When exercising in the heat, muscles require oxygen to produce energy.
b. Repeated heat stress results in acclimatization in about 10 days of exposure.
c. Concentrated carbohydrate drinks impair gastric emptying and slow fluid replacement.
d. **A hot, humid environment increases the evaporative cooling component**.

304. Fluid replacement or fluid resuscitation is the medical practice of replenishing bodily fluid lost through sweating, bleeding, fluid shifts or other pathologic processes. Fluids can be replaced via oral administration (drinking), intravenous administration, rectally, or hypodermoclysis, the direct injection of fluid into the subcutaneous tissue. Fluids administered by the oral and hypodermic routes are absorbed more slowly than those given intravenously. Which statement is incorrect in terms of fluid replacement?

a. Maintain plasma volume.
b. Colder fluids are emptied from the stomach more rapidly than room temperature fluids.
c. Concentrated carbohydrate drinks impair gastric emptying and slow fluid replacement.
d. **fluids cannot be injected directly into areas of muscle**.

305. Obesity is a medical condition in which excess body fat has accumulated to the extent that it may have an adverse effect on health, leading to reduced life expectancy. Which statement is correct in terms of obesity?

a. Excess fat decreases the metabolic cost of activity.
b. Lesser fat slows conduction of heat to the periphery.
c. Cardiac output is better regulated
d. **Activity can compromise thermoregulation in the obese with potentially fatal results**.

306. Flexibility, mobility and suppleness all mean the range of limb movement around joints. The objective of flexibility training is to improve the range of movement of the antagonistic muscles. Which statement is controversial in terms of flexibility?

a. Dynamic flexibility refers to the active ROM of a joint and is dependent upon the amount of tissue resistance met during active movement.
b. The musculotendinous unit elongates as the body segment moves through the ROM.
c. **Static stretching uses the momentum of a moving body or a limb in an attempt to force it beyond its normal range of motion**.
d. Passive flexibility is the degree to which a joint can be passively moved through the available ROM.

307. Stretching involves any therapeutic technique that lengthens shortened soft tissue structures and increases ROM. Which statement is incorrect in terms of stretching?

Type of stretching is determined by the type of force applied, the intensity of stretch, and duration of stretch.

a. Ballistic stretching uses the momentum of a moving body or a limb in an attempt to force it beyond its normal range of motion.
b. Isometric stretching involves the assistance of a partner who must fully understand what their role is otherwise the risk of injury is high.
c. Dynamic stretching consists of controlled leg and arm swings that take you gently to the limits of your range of motion.

308. Manual passive stretching takes the structures beyond the free ROM to elongate tissues beyond their resting length. Keeping this into mind which statement is incorrect in terms of manual passive stretching?

a. The stretch force is applied for at least 1530 seconds and repeated several times during a session.
b. Manual stretching is considered a short duration stretch, maintained statically for less time.
c. Intensity and duration depend on patient tolerance and therapist strength and endurance.
d. High intensity manual stretch, applied as long as possible is better.

309. Ballistic stretching is a high-intensity very short-duration "bouncing" stretch. By contracting the opposite muscle group, the patient uses body weight and momentum to elongate the tight muscle. Which statement falsely justifies ballistic stretching?

a. Facilitates the stretch reflex, causing an increase in tension in the muscle that is being stretched.
b. It should not be performed after an injury or surgery.
c. **The key to using ballistic stretching properly is to cool down with mild aerobic activity and vigorous bouncing motion**.
d. Employs a repetitive bouncing motion to induce a stretch.

310. Prolonged mechanical stretching is a low intensity external force (5 to 15 lb. to 10 percent of body weight) applied over a prolonged period by positioning a patient with weighted pulley and traction systems. Which statement falsely justifies prolonged mechanical stretching?

a. **The Prolonged Mechanical stretching devices apply a very high intensity stretch force**.
b. Dynamic splints are applied for 8 to 10 hours to increase ROM.
c. Prolonged stretch may be maintained for 20 to 30 minutes or as long as several hours.
d. Low-intensity prolonged mechanical stretching has been shown to be more effective than manual passive stretching with long-standing flexion contractures.

311. Active stretching occurs when voluntary, unassisted movement by the patient provides the stretch force to a joint. It requires strength and muscular contraction of the prime mover to actively stretch the antagonist muscle group. Which statement is incorrect in terms of active stretching?

a. The force is controlled by the patient and is considered low-intensity.
b. **Duration is more than passive manual stretching**.
c. Active stretching increases active flexibility and strengthens the agonistic muscles.
d. Many of the movements (or stretches) found in various forms of yoga are active stretches.

312. Contractile tissue is nothing more than the tissue used in muscles. It is so called because the tissue is able to contract, either on demand or involuntary. Contractile tissue has a number of unique properties in order to make it contract and exert force. Which is the false statement about contractile tissue?

a. A muscle immobilized in a shortened position will have a decrease in the number of sarcomeres and an increase in connective tissue.
b. The sarcomere adaptation is transient.
c. **The body removes the calcium from the tissue and the troponin-tropomyosin molecules, allowing returning to their normal shape**.
d. A muscle that is lengthened over a prolonged period of time will have an increase in the number of sarcomeres in series. The muscle will adjust its length over time.

313. Tissue is a cellular organizational level intermediate between cells and a complete organism. Which is not a neurophysiologic property of contractile tissue?

a. Fast stretching especially applied at end range.
b. The Golgi tendon organ (GTO) inhibits contraction of the muscle.
c. A quick stretch to a muscle stimulates the alpha motoneurons and facilitates muscle contraction via the monosynaptic stretch reflex.
d. The muscle spindle monitors the velocity and length changes in muscle.

314 Non-contractile connective tissue including ligaments, tendons, joint capsules, fasciae and skin, can affect joint flexibility and requires remodeling to increase length. Which is the incorrect statement related to Non-contractile connective tissue?

a. Intensive stretching is usually not done every day in order to allow time for healing.
b. With aging, collagen loses its elasticity and tissue blood supply is decreased reducing healing capability.
c. High-magnitude loads over long periods increase the deformation of noncontractile tissue.
d. 15 to 20 minutes of low-intensity sustained stretch, repeated on 5 consecutive days, can cause a change in the length of muscle and connective tissue.

315. Contracture is the adaptive shortening of muscle or other soft tissues that cross a joint; contracture results in decreased ROM. Which statement is incorrect in terms of contracture?

a. Scar tissue adhesions develop in response to injury and the inflammatory response.
b. Irreversible contracture is a permanent loss of soft tissue extensibility that cannot be released by any surgical treatment.
c. Myotatic contracture (pertaining to muscle) involves a musculotendinous unit that adaptively increases with gain of ROM.
d. Adhesions can occur if tissue is immobilized in a shortened position for extended periods of time, resulting in a loss of mobility.

316. Relaxation of Muscles is a technique for reducing anxiety by alternately tensing and relaxing the muscles. Which statement is incorrect in terms of relaxation of muscles in heat?

a. The GTO sensitivity is increased which makes it more likely to fire and inhibit muscle tension.
b. **Heat without stretching has maximum effect on long-term improvement in muscle flexibility**.
c. Low-intensity active exercise performed prior to stretching will increase circulation.
d. Massage increases local circulation to the muscle and reduces muscle spasm and stiffness.

317. Flexibility training, or stretching, is probably the most neglected of all the components of a fitness program. Which is not a common error associated with mobility and flexibility training?

a. Passively forcing a joint beyond its normal ROM.
b. Aggressively stretching a patient with a newly united fracture or osteoporosis may result in fracture.
c. Overstretching of weak muscles, especially postural muscles that support the body against gravity.
d. **Low-intensity, Long-duration stretching procedures on muscles**.

318. Proximal segments and trunk provide a stable base for functional movements. Which is a false statement related to dynamic stabilization, controlled mobility?

a. Distal segments are fixed while proximal segments are moving.
b. Movement normally occurs through increments of range.
c. **An individual maintains postural stability of the trunk**.
d. An individual maintains postural stability of the trunk while weight shifting.

319. It is important to consider exercise protocols that effectively challenge core muscle groups and create adequate stability to perform functional activities. Which is not a guideline to develop postural stability?

a. Start training by teaching safe spinal ROM in a variety of basic postures. Teach chin tucking with axial extension.
b. Incorporate procedures to retrain kinesthetic awareness of postural position.
c. Incorporate procedures to retrain kinesthetic awareness of postural position.
d. The lower the center-of mass and larger the base-of-support, the greater the degree of postural challenge.

320. There are certain procedures to retrain kinesthetic awareness of postural position in order to teach the neutral pelvis position first to ensure a stable base. So,which of the following procedure is not appropriate for teaching the neutral pelvis position first to ensure a stable base?

a. Visual, verbal, and proprioceptive cues.
b. Focus on patient awareness of normal alignment of the spine and pelvis feel of the muscles contracting to maintain that position while exercising.
c. Emphasis is placed on strength and endurance of back multifidi and oblique abdominals rather than erector spinae.
d. Insist on Rhythmic Stabilization

321. Alternating isometric contractions between antagonists can enhance stabilizing contractions and develop postural control e.g., PNF techniques which of the following technique do not come under the sub head of PNF stretching?

a. Rhythmic Initiation
b. Rhythmic Stabilization
c. Hold Relax.
d. **Kitchens sink exercises**.

322. The word "posture" comes from the Latin verb "ponere" which is defined as "to put or place. Which is not a common error associated with postural stability training?

a. Progressing too quickly or starting at too high a functional level for the patient to maintain postural stability.
b. Exercising past the point of fatigue, this is determined by the inability of the trunk.
c. Inadequate stretching of tight muscles.
d. Tight control of core muscles could place minimum stress on proximal structures.

323. An exercise ball is a ball constructed of elastic soft PVC with a diameter of approximately 35 to 85 centimeters (14 to 34 inches) and filled with air. The air pressure is changed by removing a valve stem and either filling with air or letting the ball deflate. It is most often used in physical therapy. Which of the following is not an advantage of stability ball training?

a. Promotes balance provides an unstable base of support, requiring continuous adjustments in balance.
b. Allows a safe, dynamic cardiovascular workout.
c. **Used to increase tone in hypertonic patient**.
d. Improves range of motion, allows safe stretching.

324. The knee joint joins the thigh with the leg and consists of two articulations: one between the femur and tibia, and one between the femur and patella. Which factor is not important in determining appropriate ball size?

a. Supine with ball under knees, the ball height should equal the distance between the greater trochanter and the knee.
b. Sitting on ball with feet flat, the ball height should place the hips and knees at 90 degree angles.
c. Movement of feet.
d. Safe practice of falling.

325. Joint stability refers to the resistance offered by various musculoskeletal tissues that surround a skeletal joint. Which is not a necessary precaution need to be taken care in stability ball training?

a. Obese individuals, exceeding ball weight limits.
b. **Ataxic patients with postural instability**.
c. Requires adequate space around exercising individual
d. Increased pain with mobility exercises and degenerative joint disease.

326. Balancing requires concurrent processing of inputs from multiple senses, including equilibrioception (from the vestibular system), vision, and perception of pressure and proprioception. Which is a goal and outcome of coordination and balance training among the following:

a. Aerobic capacity and endurance gets diffused.
b. **Performance, independence, and safety are Performance, independence, and safety is improved in transfers**.
c. Motor function gets interrupted.
d. Lack of Self-management symptoms.

327. A balance board—which is an unstable standing surface that moves in different planes of motion—is commonly used to train balance. Which is not a training strategies to improve coordination and balance among the following?

a. Sensory cues are used to enhance motor performance.
b. Promote adaptability and generalize ability of skills.
c. Patient decision making skills are promoted.
d. **Feedback given gradually improves initial performance**.

328. Compensatory strategies are utilized as appropriate to promote safety and early resumption of functional skills, e.g., the patient with delayed or absent recovery, multiple co-morbidities. Compensatory training may lead to learned nonuse of impaired extremities and delay recovery in those patients with recovery potential. Which of the following is an incorrect statement relating to compensatory strategies?

a. Safety is improved by use of appropriate assistive devices and shoes.
b. **Safety cannot be improved by substitution**.
c. Safety is improved through environmental adaptations.
d. Safety is improved by altering postural strategies.

329. Remediation/facilitation approaches reduce the effects of specific impairments; focus is on use of involved body segments (e.g., affected extremities in the patient with stroke).

Which of the following is an incorrect statement related to remediation?

Control is first achieved in holding (stability) before moving in a posture.

a. As quality of movement improves, speed of movement and control is increased.
b. **Control is always developed in complex situations to focus on various body skills**.
c. Specific techniques can be used to remediate impairments.

330. Functional training is any type of exercise that has a direct relationship to the activities you perform in your daily life. Which one of the following is not an intervention to improve coordination in functional training?

a. Environment: patients with ataxia do better in a low stimulus environment; allows better utilization of cognitive strategies.
b. Specific exercise techniques to enhance stability
c. Progress to controlled mobility activities: weight shifting through increment
d. Use slow-reversal-hold (SRH) through decreasing ROM with ataxic movements.

331. Sensory Training is mostly used for Patients with proprioceptive losses .Which one of the following statements is incorrect in terms of sensory training?

a. Light weights like wrist cuffs, ankle cuffs, weighted walkers, elastic resistance bands to increase proprioceptive loading.

b. **Stabilization devices are used to overcome.**

c. Visual compensation strategies are adopted.

d. Patients with visual losses benefit from cognitive training strategies along with environmental adaptations and assistive devices.

332. When exercising the ability to balance, one is said to be balancing. Which of the following is not an intervention to improve balance among the following?

a. **Floor-to-standing rises**

b. Training of change-of-support strategies.

c. Weight shifts

d. Postural awareness training.

333. Functional training activities are a classification of exercise which involves training the body for the activities performed in daily life. Functional training may lead to better muscular balance and joint stability. Which of the following is not a functional training activity among the following?

a. Sit-to-stand (STS) and sit-down (SIT) activities.
b. Elevation activities
c. Gait activities
d. **Kitchen sinks exercises**.

334. Balance training is a type of exercise which is designed to improve balance and proprioception, the sensation of knowing where the body and its joints are in space. Which of the following is not a disturbed balance activity among the following?

a. Therapist-initiated manual perturbations.
b. Stability ball training
c. Dual-task training
d. Wobble board/equilibrium boards

335. A sensory system is a part of the nervous system responsible for processing sensory information. Which change is incorrect due to the effects of sensory changes?

a. Somatosensory changes
b. Visual changes
c. Vestibular changes
d. Gait activities

336. In balance training, the goal is to increase the body's agility, and to get someone in touch with his or her center of gravity. Which one of the following is a false prevention to improve balance?

a. Assist patient in identification of fall risk factors.
b. Lifestyle counseling
c. Assist the patient about the harmful effects of a sedentary lifestyle.
d. **Assist in monitoring adequate nutritional intake**.

337. Aerobic endurance training improves aerobic capacity by 5% to 25% in previously untrained, healthy adults. The magnitude of improvement is primarily dependent upon the initial level of physical fitness. Which is a false intervention to improve aerobic capacity and endurance among the following?

a. Pace pedaling on a cycle ergometer; progression is from slow to fast.
b. Treadmill walking: focus on velocity control; progression is from slow to fast. Safety harness can be worn to provide partial body weight support (BWS) if patient is unstable
c. Stretching exercises
d. **Manual resistance like PNF patterns cannot be used**.

338: Homeostasis regulated through thirst mechanisms and renal function via circulating antidiuretic hormone (ADH). Which of the following is called hypokalemia?

A) an excess of body fluids with expansion of interstitial fluid volume.
B) excessive loss of body fluids or fluid output exceeds fluid intake.
C) **decreased potassium in the blood**.
D) excess of potassium in the blood; common in acute renal failure.

339. Wound debridement is removal of necrotic or infected tissue that interferes with wound healing. Which of the following is true with respect to wound debridement:

a. Allows examination of ulcer, determination of extent of wound
b. Decreases bacterial concentration in wound;
c. Improves wound healing.
d. **Increases spread of infection, i.e., cellulitis or sepsis**.

340. Electrical stimulation is a treatment modality utilized by physical therapists that uses an electrical current to cause a single muscle or a group of muscles to contract. This contraction helps strengthen injured muscles and promotes healing. Which of the following is false with respect to electrical stimulation:

a. Uses capacitive coupled electrical current to transfer energy to a wound, improve circulation, facilitate debridement, enhance tissue repair.
b. Continuous waveform application with indirect current
c. High-voltage pulsed current (HVPC)
d. Microcurrent electrical stimulation (MENS)

341. Nutrition is the provision, to cells and organisms, of the materials necessary (in the form of food) to support life. Many common health problems can be prevented or alleviated with a healthy diet. Which of the following is false with respect to nutritional considerations?

a. Delayed wound healing associated with malnutrition and poor hydration.
b. Provide adequate hydration: twelve 12-oz glasses of non-caffeine fluids per day unless contraindicated.
c. Provide adequate nutrition: frequent high calorie/ high-protein meals
d. Patients with trauma stress and burns require higher intake.

342. A therapeutic effect is a consequence of a medical treatment of any kind, the results of which are judged to be desirable and beneficial. This is true whether the result was expected, unexpected, or even an unintended consequence of the treatment. An adverse effect, on the other hand, is a harmful and undesired effect. Which of the following is true with respect to therapeutic positioning to relieve pressure and tissue reperfusion in case of injury prevention or reduction?

a. In bed: turning or repositioning schedule every 2 hours during acute and rehabilitation phases.
b. In bed: turning or repositioning schedule every 5 hours during acute and rehabilitation phases.
c. In wheelchair: wheelchair push-ups every 50 minutes.
d. In wheelchair: wheelchair push-ups every 5 hours.

343. In dermatology, an abrasion is a wound caused by superficial damage to the skin, no deeper than the epidermis. Which of the following is not a unique technique to ensure skin protection, avoid friction, shear or abrasion injury?

a. Use of cornstarch, lubricants, pad protectors, thin film dressings, or hydrocolloid dressings over friction risk sites
b. Use of turning and draw sheets; trapeze, manual or electric lifts.
c. Dragging, not lifting
d. Use of transfer boards for sliding wheelchair transfers.

344. The NICE clinical guideline on pressure relieving devices makes recommendations about how the risk of developing a pressure ulcer can be assessed and how pressure ulcers can be prevented by using devices designed to reduce pressure. Which of the following is a correct explanation of the given type of pressure-relieving device?

a. Static devices: use if patient cannot assume a variety of positions; examples include alternating pressure air mattresses, fluidized air or high-air-loss bed.
b. Dynamic devices: use if patient can assume a variety of positions; examples include foam, air, or gel mattress overlays; water-filled mattresses; pillows or foam wedges, protective padding (heel relief boots)
c. **Seating supports: use for chair-bound or wheelchair-bound patients; examples include cushions made out of foam, gel, air, or some combination**.
d. Seating supports: use if patient cannot assume a variety of positions; examples include alternating pressure air mattresses, fluidized air or high-air-loss bed.

345. Maceration refers to skin changes seen when moisture is trapped against the skin for a prolonged period. The skin will turn white or gray, softens and wrinkles. Macerated skin is more permeable and prone to damage from friction and irritants. Which of the following is false for avoiding maceration injury?

a. Prevent moisture accumulation
b. Temperature elevation where skin contacts support surface
c. Use of absorbent pads, brief or panty pad
d. Do not use ointments, creams, and skin barriers prophylactically in perineal and perianal areas

346. An immune system is a system of biological structures and processes within an organism that protects against disease by identifying and killing pathogens and tumor cells. Which of the following is not included in the immune system?

a. immune cells
b. central immune structures where immune cells are produced
c. peripheral immune structures
d. HBOT (Hyperbaric oxygen therapy)

347. An immune system consists of immune cells. There are several different types of immune cells. Which of the following is an incorrect explanation of immune cells?

a. Antigens (immunogen) are molecules that link immune cells with other tissues and organs.
b. Lymphocytes (T and B lymphocytes) are the primary cells of the immune system.
c. CD molecules (e.g. CD4 helper cells) serve as master regulators of the immune response by influencing the function of all other immune cells.
d. Recognition of foreign threat from self (autoimmune responses) is mediated by MHC membrane molecules.

348**.** The thymus is the primary central gland of the immune system. It is located behind the sternum above the heart and extends into the neck region to the lower edge of the thyroid gland. Which of the following is false with respect to thymus?

a. It is fully developed at birth and reaches maximum size at puberty.
b. It further increases in size
c. It is slowly replaced by adipose tissue
d. It produces mature T lymphocytes

349. You are evaluating an athlete who is complaining of pain in the left shoulder region. Your examination of the shoulder elicits pain in the last 30 degrees of shoulder abduction range of motion. This finding is most congruent with:

A. Calcific supraspinatus tendinitis.

B. Subacromial bursitis.

C. **Acromioclavicular sprain**.

D. Thoracic outlet syndrome.

350. Following a cerebrovascular accident involving the dominant right hemisphere, a patient exhibiting unilateral neglect would generally not:

A. Eat food only from the right side of a plate.

B. Bump a one-arm driven wheelchair into things on the left side.

C. Ignore or deny the existence of the left-sided limbs.

D. Shave or put makeup only on the left side of the face.

351. To help decrease shear when transferring a patient with a spinal cord injury from bed to a chair it would be best to use:

A. **A draw sheet**.

B. Skin lubricant on involved skin surfaces.

C. An air mattress.

D. A sheepskin pad.

352. Prior to performing manual traction in the cervical region one should perform a test for vertebral artery insufficiency in order to ascertain the adequacy of blood supply to the:

A. Vertebrae.

B. Upper extremities.

C. Spinal canal.

D. Brain stem.

353. To prepare a patient with a cauda equina lesion for ambulation with crutches, the upper quadrant muscles that would be the most important to strengthen would be the:

A. Upper trapezius, rhomboids and levator scapulae.

B. Deltoid, coracobrachialis and brachialis.

C. Middle trapezius, serratus anterior and triceps.

D. **Lower trapezius, latissimus dorsi and pectoralis major**.

354. Which activity would not help to break up lower extremity synergy combinations in a patient with hemiplegia?

A. Balance training in a kneeling position.

B. Backward leg lifts with the knee extended on the affected side.

C. Bridging in the hooklying position.

D. Rolling from a hooklying position.

355. Following a reattachment of the flexor tendons of the fingers, one physical therapy goal is to minimize adhesion formation. A few days after surgery, with the patient in a splint, the physical therapist should teach the patient to perform:

A. Passive extension and active flexion of the interphalangeal joints.

B. Active extension and flexion of the interphalangeal joints.

C. Active extension and passive flexion of the interphalangeal joints.

D. Gentle passive extension and flexion of the interphalangeal joints.

356. The most functional way to teach an individual with a T4 complete paraplegia to transfer from wheelchair to mat is by using a:

A. A stand pivot technique.

B. **A squat pivot technique**.

C. Sliding board.

D. Back-out technique.

357. Your plan of care includes use of iontophoresis in the management of calcific bursitis of the shoulder. To administer this treatment using the acetate ion, the current characteristics and polarity should be:

A. Monophasic twin peaked pulses using the positive pole.

B. Monophasic twin peaked pulses using the negative pole.

C. Continuous monophasic using the positive pole.

D. Continuous monophasic using the negative pole.

358. Following a mastectomy, a sixty-three year-old female developed massive edema of the arm on the involved side. A compression garment was ordered to help the situation. To help decrease the edema, this garment must exert enough pressure to:

A. Decrease the osmotic pressure of the capillaries.

B. Increase the capillary permeability.

C. **Exceed the internal tissue hydrostatic pressure**.

D. Equal the fluid outflow from the capillaries.

359. If you are treating a patient with active infectious Hepatitis B, transmission of the disease is best minimized if you take precautions to avoid:

A. **Direct contact with the patient's blood or blood-contaminated equipment**.

B. Direct contact with any part of the patient.

C. Droplet spread of the organisms by coughing.

D. Direct contact with the patient's hands.

360. You are working in a school setting as a physical therapist. A supervisor asks you to organize a scoliosis screening program. It would be best to assess:

A. 2^{nd} or 3^{rd} graders.

B. **6^{th} or 7^{th} graders**.

C. High school freshmen or sophomores.

D. High school juniors or seniors.

361. Use of continuous ultrasound at 1.5 watts/cm2 will result in:

A. No change in nerve conduction velocity.

B. Increase in motor nerve conduction velocity and decrease in sensory nerve conduction velocity.

C. Decrease in motor nerve conduction velocity and increase in sensory conduction velocity.

D. Increase in both motor and sensory never conduction velocity.

362. It is most likely that when treating a patient with Lyme Disease of more than one year's duration, the physical therapy focus will be on management of arthritic changes primarily affecting the:

A. Small joints of the hands and feet.

B. **Large joints of the body, especially the knee**.

C. Axial joints, especially the lumbrosacral spine.

D. Axial joints, especially the cervical and thoracic spine.

363. While gait training a patient following a cerebral vascular, you observe the knee on the affected side going into recurvatum during stance phase. The most likely cause of this deviation can be attributed to:

A. Severe spasticity of the hamstrings or weakness of the gastrocnemius-soleus.

B. Weakness or severe spasticity of the quadriceps.

C. Weakness of the gastrocnemius-soleus or spasticity of the pretibial muscles.

D. Weakness of both the gastrocnemius-soleus and pretibial muscles.

364. You are organizing a group exercise session in a therapeutic pool. Use of the pool would be contraindicated for a group member who has:

A. An indwelling catheter.

B. Paraparesis and is on a regular bowel program.

C. **Unstable blood pressure**.

D. An open skin lesion, even if it is covered by a waterproof dressing.

365. To prevent contractures in a newly admitted 8 year-old with anterior neck bums, it would be best to position the neck in:

A. Hyper flexion.

B. Slight flexion.

C. Neutral.

D. Extension.

366. You are treating a 54 year-old woman for degenerative arthritis of the left knee. Her medical record indicates that she is on estrogen replacement therapy. In this case, you should consider that this patient might be more susceptible to:

A. Coronary artery disease.

B. Osteoporosis.

C. Clinical depression.

D. Weight gain.

367. You are instructing a new mother to perform range of motion and stretching on her newborn who has a club foot. You would advise her to carefully stretch in the direction of:

A. Plantar flexion and inversion.

B. Plantar flexion and eversion.

C. Dorsiflexion and inversion.

D. **Dorsiflexion and eversion**.

368. The normal end-feel associated with full elbow extension can be classified as:

A. Empty.

B. Capsular.

C. Springy block.

369. **Bone-on-bone**.

370. If a patient had normal quadriceps strength; but, unilateral weakness (3/5) of the hamstring muscles, during swing phase you might observe:

A. Excessive compensatory hip extension on the sound side.

B. Decreased hip flexion followed by increased knee flexion on the weak side.

C. Excessive hip extension followed by abrupt knee extension on the weak side.

D. **Excessive hip flexion followed by abrupt knee extension on the weak side**.

371. A patient wishes to improve her aerobic fitness. She currently jogs four days a week for 30 minutes at 70% of her age-predicted maximum heart rate. The recommendation that would not result in improved aerobic fitness is:

A. Increasing the distance covered in the same 30 minutes.

B. Increasing the jogging time to 45 minutes while keeping at 70% of the age-predicted heart rate.

C. **Changing to interval training with maximum burst of running for 15 seconds, followed by a 30 second rest. Complete 4 sets per day, 4 days per week**.

D. Changing to interval training for 4 days per week by doing 90 seconds of comfortable running followed by 90 seconds of rest for a period of 30 minutes.

372. Four days following open heart surgery, a patient is being treated in the physical therapy department. He complains of some chest discomfort during treatment and wishes to return to his room. You should:

A. Call his physician immediately.

B. Complete the treatment and have an aide transport him back to his room as some discomfort is expected.

C. Call the nurse and check to see if the discomfort is to be expected.

D. **Immediately transport the patient back to his room yourself and inform nursing services of the patient's complaint**.

373. A patient who is to undergo surgery for a chronic shoulder dislocation asks you to explain the advantages and disadvantages to the various surgical reconstructive procedures used to alleviate the problem. Your best response is to:

A. Give the patient as much information about the procedures as you currently know.

B. Explain how patients you have treated responded to the surgery.

C. Tell the patient to ask the surgeon for this information since this precise information is outside the scope of physical therapy practice.

D. Refer the patient to another therapist in the department who is an expert on shoulder reconstructive rehabilitation.

374. A patient's peripheral skin color progresses from blue to white to red. This would be most characteristic of:

A. Chronic venous insufficiency.

B. Acute venous insufficiency.

C. Acute arterial insufficiency.

D. Vasomotor disorders.

375. You see a patient who had a CVA two weeks ago. The patient has motor and sensory impairments primarily in the opposite lower extremity. There is some confusion and perseveration. Based on these findings, the vascular problem can be characterized as:

A. Transient ischemic attack.

B. Internal carotid syndrome.

C. **Anterior cerebral artery syndrome**.

D. Middle cerebral artery syndrome.

376. In the management of systemic lupus erythematosus, the intervention that would not be appropriate would be the use of:

A. Nonsteroidal anti-inflammatory agents to control arthralgia.

B. Resting splints to decrease joint pain and prevent deformity.

C. Ultraviolet irradiation to help decrease the skin lesions and rash often associated with the disorder.

D. Endurance training to compensate for cardiopulmonary dysfunction that is often present with the disorder.

377. A patient with degenerative joint disease of the right hip complains of pain in the anterior hip and groin which is aggravated by weightbearing. There is decreased range of motion and capsular mobility. Right gluteus medius weakness is evident during ambulation and there is decreased tolerance of functional activities including transfers and lower extremity dressing. In the case, a capsular pattern of joint motion should be evident by restriction of hip:

A. **Flexion, abduction and internal rotation**.

B. Flexion, adduction and internal rotation.

C. Extension, abduction and external rotation.

D. Flexion, abduction and external rotation.

378. Confirmation of a diagnosis of spondylolisthesis can be made when viewing an oblique radiograph of the spine. The tell-tale finding is:

A. Posterior displacement of L5 over S1.

B. Bamboo appearance of the spine.

C. Compression of the vertebral bodies of L5 and S1.

D. **Bilateral pars interaticularis defects**.

379. When fitting a patient for adjustable axillary crutches, the measurement technique that would not be appropriate to use is:

A. Placing the patient supine and measuring from the anterior axillary fold to the bottom of the foot and adding 2 inches.

B. Placing the patient supine and measuring from the anterior axillary fold a point 6 inches lateral to the foot.

C. Subtracting 16 inches from the height of the patient.

D. Placing the patient in standing with shoes on, and the crutches placed 6 inches lateral to the foot.

380. While ambulating a patient in the parallel bars, you lose control and the patient falls hitting her head on the bar. The patient lies motionless on the floor between the bars bleeding heavily from a scalp laceration. The first thing you should do is:

A. Put thick gauze over and apply manual pressure to the scalp wound.

B. **Check for responsiveness**.

C. Call Emergency Medical Services.

D. Immediately determine the patient's heart rate and blood pressure.

381. During pregnancy, the presence of the hormone relaxin can lead to abnormal movement and pain which most frequently affect the:

A. Glenohumeral joints.

B. Hip joints.

C. Iumbrosacral joints.

D. Sacroiliac joints.

382. An occupation which requires a great deal of kneeling will be impacted the most if the diagnosis is:

A. Osgood-Schlatter Disease.

B. Plica syndrome.

C. **Acute prepatellar bursitis**.

D. iliotibial band friction syndrome.

383. Three days following a cerebral vascular accident, a patient is supine in bed. It would be best to position the upper extremity so that the:

A. **Scapula is protracted and upwardly rotated and the shoulder is abducted and externally rotated**.

B. Scapula is protracted and upwardly rotated and the shoulder is abducted and internally rotated.

C. Scapula is retracted and downwardly rotated and the shoulder is adducted and internally rotated.

D. Scapula is retracted and downwardly rotated and the shoulder is abducted and externally rotated.

384. When the ankle is forcibly inverted and plantar flexed, the ligament that is most frequently disrupted is the:

A. Deltoid.

B. **Anterior talofibular**.

C. Posterior talofibular.

D. Calcaneofibular.

385. The clinical status of a truck driver with a posterior herniated nucleus pulposus has improved if:

A. Peripheral pain increases only when lumbar extension is attempted.

B. Peripheral pain occurs only with straight leg raising.

C. **Pain centralizes with passive hyperextension of the spine**.

D. There is flattening of the lumbar lordosis.

386. When instructing the family of a 9 year-old boy with Duchenne muscular dystrophy, the main emphasis in the lower extremities should be placed on:

A. Strengthening the knee extensors and plantar flexors.

B. Strengthening the plantar flexors and stretching the hip extensors.

C. **Stretching the hip flexors and plantar flexors**.

D. Strengthening the hip flexors and knee extensors.

387. An 18 year-old female soccer player, with a Q angle in excess of 30 degrees, exhibits patello-femoral tracking problems. While playing soccer, it would be best if she wore a:

A. **Patellar stabilizing brace with a lateral buttress**.

B. Patellar stabilizing brace with a medial buttress.

C. Neoprene sleeve with a patellar cutout.

D. Derotation brace.

388. In treating a 5 year-old with spastic diplegia, as part of the program, relaxation might e accomplished by:

A. Use of rhythmic stabilization.

B. **Slow rocking**.

C. Inhibition of parasympathetic fibers.

D. Facilitation of sympathetic fibers.

389. When using ultrasound in treating chronic bursitis of the hip, the most benefit might occur if the ultrasound frequency and dosage were:

A. **1 MHz and 1.5 Watts/cm2**.

B. 1 MHz and 0.5 Watts/cm2.

C. 3 MHz and 1.5 Watts/cm2.

D. 3 MHz and 0.5 Watts/cm2.

390. When performing underwater ultrasound, the most important safety factor is that the:

A. Part being treated should not be immersed in a metal tank.

B. Transducer head keeps moving.

C. Ultrasound apparatus is connected to a ground fault interruption circuit.

D. Piezoelectric crystal is not cracked.

391. A weightlifter with hypertrophy of the scalene muscles complains of pain and paresthesia in the right upper extremity when lifting the weight overhead. It is most likely that this is a manifestation of:

A. Thoracic outlet syndrome.

B. Vertebral artery obstruction.

C. Cervical radiculitis.

D. Reflex sympathetic dystrophy.

392. When using TENS to help modulate pain, one electrode is placed on the posteromedial thigh and another on the posterolateral calf. The dermatome involved in this case would be:

A. L2.

B. L4.

C. **S2**.

D. S1.

393. A five foot tall, 37 year-old word processing technician complains of lumbar discomfort after a full day at her computer work station. To help the situation, you would not recommend that she try:

A. Use of a lumbar roll.

B. Use of an elevated foot rest.

C. **Moving her chair further away from the keyboard and screen**.

D. Taking frequent short walks during the day.

394. You are working with a 12 month-old with a developmental level of 3 months. Appropriate intervention at this time would incorporate:

A. **Pivot prone positioning**.

B. Rolling from prone to supine.

C. Rolling form supine to prone.

D. Training in unsupported sitting.

395. To exercise the middle deltoid during a home health visit, you apply a cuff weight to the wrist and instruct the patient in glenohumeral abduction. Since the patient is unable to complete the exercise, you move the weight to the elbow. This change in position may allow the patient to complete the motion because it:

A. Reduces the force arm, thereby increasing the mechanical advantage of the lever.

B. Reduces the resistance arm, thereby decreasing the mechanical advantage of the lever.

C. Reduces the force arm, thereby decreasing the mechanical advantage of the lever.

D. **Reduces the resistance arm, thereby increasing the mechanical advantage of the lever**.

396. On the first day following a cesarean delivery, a physical therapist's primary treatment responsibilities would consist of teaching the new mother:

A. Gentle partial sit ups and pelvic tilting.

B. **Breathing, coughing and pelvic floor exercises**.

C. Assisted ambulation.

D. Active calf and ankle exercises to prevent venous stasis.

397. You are performing sensory tests on a patient diagnosed with C6 nerve root impingement. Testing should concentrate on the:

A. 3^{rd}, 4^{th} and 5^{th} fingers.

B. Ulnar border of the hand.

C. **Thumb and index fingers**.

D. Medial forearm.

398. The muscles that can function as synergists when acting on the scapula are the:

A. **Rhomboids and trapezius**.

B. Rhomboids and serratus anterior.

C. Serratus anterior and levator scapulae.

D. Teres major and infraspinatus.

399. The left phrenic nerve of a patient was accidentally severed during thoracic surgery. The physical therapist should work on facilitating and strengthening all of the following muscles to provide substitute function with the exception of the:

A. Sternocleidomastoid.

B. External intercostals.

C. Scalenes.

D. **External obliques**.

400. A physician requests that you perform hydrocortisone iontophoresis over the left shoulder of a patient with tendonitis. You discover that the patient has a pacemaker. In this case:

A. Perform the treatment since there is no contraindication as long as nothing is done directly over the pacemaker.

B. Refer the patient to another physical therapist who has greater expertise in using iontophoresis.

C. **Begin to examine the patient and consult with the physician about alternate forms of therapy; however, do not perform the iontophoresis treatment**.

D. Ethically you must refuse to administer iontophoresis and have the patient return immediately to the physician.

401. A patient, in trying to cope with his disability, is going through a stage of anger. Anger would not be expressed using the mechanism of:

A. **Denial**.

B. Projection.

C. Noncompliance with treatment.

D. Displacement.

402. When having an individual perform interval training to stress the aerobic system, a work interval of three minutes is employed. It would be best if the rest interval equals:

A. One minute.

B. **Three minutes**.

C. Six minutes.

D. Nine minutes

403. You are evaluating a patient with a temporomandibular disorder. The patient had a history of jaw clicking on opening and closing the mouth, but the noises are now gone. The current major complaint is inability to open the mouth wide and difficulty performing functional jaw movements such as chewing and yawning. You suspect that the most likely cause is:

A. Capsular fibrosis.

B. TMJ subluxation.

C. Chronic TMJ osteoarthritis.

D. **Disc displacement without reduction**.

404. A 13 year-old girl has a structural right thoracic idiopathic scoliosis. The clinical features you would expect to find include:

A. **A high right shoulder, a prominent right scapula and a left hip that protrudes**.

B. A high left shoulder, a prominent left scapula and a right hip that protrudes.

C. A high right shoulder, a prominent left scapula and a right hip that protrudes.

D. A high left shoulder, a prominent tight scapula and a left hip that protrudes.

405. While evaluating a 2 year-old child with mild cerebral palsy, the therapist is encouraged because the normal developmental milestones for a child of this age have been achieved. This was demonstrated by the child's ability to:

A. Hop on one foot.

B. Stand on tiptoes.

C. **Ascend stairs reciprocally**.

D. Begin to self-feed.

406. Gryotherapy should be avoided with patients exhibiting:

A. Myofascial pain syndrome.

B. Severe spasticity.

C. Degenerative joint disease.

D. Vasospasm.

407. Following serious trauma, a patient is casted as a result of multiple bilateral wrist and hand fractures. It would be best if physical therapy intervention begins:

A. **As early as possible to maintain or regain strength and range of motion in nonimmobilized joints of the upper extremities**.

B. As soon as the casts are removed to regain strength of the wrists and hands.

C. About two weeks after the casts are removed so as not to damage vulnerable soft tissue.

D. After six weeks of immobilization to ensure that bone healing is almost complete.

408. Your patient with flaccid hemiplegia exhibits pain in the shoulder region secondary to glenohumeral subluxation. Using electrical stimulation as orthotic substitution, it would be best to place the electrodes over the:

A. Supraspinatus and upper trapezius.

B. Supraspinatus and posterior deltoid.

C. Anterior and posterior deltoid.

D. Anterior, middle and posterior deltoid.

409. Increasing the frequency of ultrasound in order to treat an ankle sprain will result in:

A. **Increased attenuation**.

B. Decreased tissue temperature.

C. Decreased intensity.

D. Increased intensity.

410. A patient has osteophyte formation affecting the atlanto-axial joint with resultant cervical nerve root irritation. If the physical therapist elects to use gentle static manual traction to help deal with this problem, it would be best if the neck was positioned in:

A. 10-15 degrees of hyperextension.

B. **Neutral or 0 degrees of flexion**.

C. 20 degrees of flexion.

D. 30 degrees of flexion

411. A realistic functional outcome for a patient with a complete lesion at the C8 neurological level is independence in:

A. Transfers using a sliding board.

B. Using a manual wheelchair with rim projections.

C. **All self-care and personal hygiene**.

D. Driving an automobile without hand controls.

412. A patient with a crush injury to the foot developed complex regional pain syndrome (CRPS). Now, two mothers into the CRPS, the clinical presentation you would expect is:

A. Edema and osteoporosis with decreased sweating and nail growth.

B. A cool, dry extremity with the beginnings of ankylosis.

C. Causalgia with vasomotor reflex spasm resulting in warm, dry skin with increased nail growth.

D. Pain on motion with trophic skin changes and osteoporosis.

412. What condition commonly results in bone spurring?

a) tendinitis.

b) migraines.

c) **osteoarthritis.**

d) neuritis.

413. Typical head position for a client with the right SCM in contracture from Torticollis would be:

a)lateral flexion to the left, rotation to the left **.**

b)lateral flexion to the right, rotation to the right .

c) **lateral flexion to the right, rotation to the left.**

d) lateral flexion to the left, rotation to the right.

414. Rotator cuff tendinitis does not include which tow tendon areas?

a)**deltoid and teres major .**

b) infraspinatus and subscapularis.

c) supraspinatus and teres minor.

d) supraspinatus and subscapularis.

415. The painful arc in active shoulder abduction occurs at approximately:

a) **50 to 120 degrees.**

b) 180 degrees.

c) when the arm comes down.

d) The first 10 degrees.

416. The pain in a painful arc test is the result of:

a)bracial plexus impningement.

b) pressure on the sternum.

c) **pressure on the inflamed rotator cuff tissue as it passes under the acrominon process.**

d) the movement of the scapula over the ribs.

417. Which of the following conditions is the result of an inflamed nerve ?

a**) neuritis.**

b) tendinitis.

c) bursitis.

d) rheumatoid arthritis.

418. Upon postural assessment, you find that the client's right foot is severely pronated. On which part of the right foot does he stand?

a)**medial .**

b) lateral.

c) on his outside heel.

d) on his toes.

419. Which muscles would you attempt to lengthen in question 7?

a) tibialis posterior and Sartorius.

b)**peroneous longus and extensor digitorum longus.**

c) tibialis posterior and gastrocnemius.

d) all of the above.

420. Thoracic outlet syndrome is a type of

a)arthritis.

b) fibromyalgia.

c) **neuralgia.**

d) chronic fatigue.

421. The borders for the posterior triangle of the neck are:

a) transverse processes of the cervicals, splenius capitus, trapezius

b) **SCM, trapezius, clavicle**

c) SCM spine of the scapula, trapezius

d) SCM, trachea, mandible

422. What type of hydrotherapy application includes active movement following numbness?

a) foot bath

b**) cryokinetics**

c) hot compress

d) vascular flush

423. Why is tendinitis slow in the healing process?

a) poor synovial fluid secretion.

b) **poor vascularization**.

c) poor mobility.

d) poor muscular support.

424. Which joint is involved with a shoulder separation?

a) sterno-clavicular.

b) **acromio-clavicular**.

c)costo-clavicular.

d) glenohumeral.

425. Which joint is involved with shoulder dislocation?

a) sterno-calvicular.

b) acromio-clavicular.

c) scapular thoracic .

d) **glenohumeral.**

426. Which of the following is defined as nerve pain?

a)neuritis.

b) tendinitis.

c) bursitis.

d**) neuralgia**.

427. A protruding 4th lumbar disc would most likely cause:

a) arthritis.

b)tendinitis.

c) bursitis.

d) **sciatica.**

428. Which ROM is indicated when assessing an acute whiplash injury?

a) **active.**

b) passive.

c) resisted.

d) assisted.

429. What would be the most appropriate massage treatment for a muscle spasm?

a) gliding friction.

b) effleurage.

c) **direct pressure**.

d) tapotement.

430. Another name for tennis elbow is:

a) thoracic outlet syndrome.

b) golfer's elbow.

c) medial epicondylitis.

d) **lateral epicondylitis**.

431. The iliopsoas hikes up the hip because of its insertion on the

a) femur.

b) greater trochanter.

c) **lesser trochanter .**

d) ischial tuberosity.

432. Which statement is true about Golgi tendon apparatus?

a) they are found in joint capsules.

b**) they detect the overall tension of the tendon.**

c) there are a higher number of them in gymnasts.

d) they insert on the greater tuberosity.

433. Which muscle is closet to the sciatic nerve?

a) gracilis.

b) **piriformis.**

c) gluteus medius.

d) pectineus.

434. The primary flexor of the distal phalanx of the fingers is

a)flexor carpi ulnaris.

b) pollices longus.

c) **flexor digitorum profundus.**

d) flexor carpi radialis.

435. Which muscle is innervated by the axillary nerve?

a) **deltoid.**

b)bracialis.

c) pectoralis major.

d) all of the above.

436. What muscle plantarflexes and everts the foot?

a) tibialis anterior.

b) gastrocnemius.

c)plantaris.

d)**peroneus longus**.

437. Which Best describes the effect of massage therapy?

a) **Increased venous and lymph flow.**

b) increased venous, decreased arterial flow .

c) decreased venous and lymph flow.

d) decreased venous increased lymph flow.

438. A pregnant client should have pillows under her back when she is lying

a) on her side.

b) **supine.**

c) prone.

d) upside down.

439. Petrissage is best used for

a) breaking up scar tissue.

b) **relieving chronic or pitted edema.**

c) relaxing tight muscles.

d) decreasing circulation e. shortening a muscle that is overstretched.

440. Development of scar tissue occurs during which stage of inflammation?

a) acute.

b) sub-acute.

c) sub-chronic.

d) **chronic.**

441. How would you test for a tight piriformis?

a) client supine, straight leg lift.

b) **client prone, internal rotation of thigh.**

c) client prone, external rotation of thigh.

d) client supine, flexion of the knee.

442. Neck ligament injuries may result in which of the following symptoms?

a) pain in the arm.

b) headaches.

c) dizziness.

d) **all of the above.**

443. Torticollis involves which muscle?

a) splenius capitis.

b) rectus capitis.

c)**sternocleidomastoid.**

d) rectus longus colli.

444. Improperly positioned wheelchair leg and footrests could result in ____?

A. Excessive pressure under the distal thigh .
B. Excessive pressure under the ischial tuberosities .
C. Contractures of the lower extremities .
D. **All of the above** .

445. When performing goniometric measurements a joint should be placed in the anatomical position . Which joint motions are exceptions to this rule?
A. Forearm supination and pronation .
B. Hip rotation .
C. Shoulder rotation .
D. All of the above .

445. Which type of electrical current would most likely select if your treatment objective is to stimulate denervated muscle ?

A. Alternating current .
B. **Direct current** .
C. Pulsatile current .
D. Denervated muscle responds in a similar manner to all types of current .

447. which of the following situations would be contraindicated when using transcutaneous electrical nerve stimulation ?

A. Use over an arthritic joint .
B. Use during labor and delivery .
C. **Use over a pregnant uterus**.
D. Use to diminish phantom limb sensation .

448. A manual muscle test is conducted on the medial portion of the deltoid . The muscle is able to move through the full range of motion with gravity eliminated ,but can not function against gravity . The results of the manual muscle test should be recorded as ____?

A. Trace .
B. Poor.
C. Fair.
D. Good.

449. Iontophoresis is used to treat all of the following conditions except____?

A. Muscle spasms .
B. Ischemic skin ulcers .
C. Fungal infection .
D. Bursitis and tendonitis .

450. A therapist measures passive forearm pronation and concludes the results are within normal limits . Which measurement would be classified as within normal limits?

A. 65 degrees .
B. **85 degrees** .
C. 105degrees .
D. 125 degrees .

451. Effective measures to prevent decubitus ulcers include all of the following except _______?

A. Frequent inspection of the skin for redness and signs of skin breakdown .
B. **Turning and position changes every 12 hours** .
C. Early ambulation when possible .
D. Passive range of motion exercises .

452. A patient has acute synovitis of the left temporomandibular joint. Mandibular depression is limited.You would also expect to find:

a) left lateral excursion to be less than functional.
b) **right lateral excursion less than functional**.
c) protrusion to be functional.
d) protrusion to be less than functional with deflection to the right.

453. With the patient sitting on a therapy ball, training and strengthening the lumbar extensors is best accomplished by:

a) rolling the ball in a circular pattern.
b) rolling the ball backward.
c) pushing the pelvis backward.
d) **pushing the pelvis forward**.

454. A patient has loss of the last 15 degrees of elbow extension. To regain motion the PT should implement:

a) volar glide of the radius on the humerus.
b) **dorsal glide of the radius on the humerus**.
c) dorsal glide at the radioulnar joint.
d) volar glide at the radioulnar joint.

455. History reveals that a patient awoke from an afternoon nap and was very unsteady on her feet. She loses her balance to the right. There is horizontal nystagmus and loss of conjugate gaze on the right with impaired sensation of the contralateral face, limbs and trunk. These clinical manifestations are most suggestive of:

a) **superior cerebellar artery syndrome**.
b) anterior inferior cerebellar artery syndrome.
c) lateral medullary syndrome.
d) medial medullary syndrome.

456. A patient has difficulty during the early swing phase of gait. She circumducts her leg to bring it through swing. A likely cause of this problem is:

a) flexor withdrawal reflex.
b) hamstring spasticity.
c**) knee pain with decreased ROM**.
d) dysmetria.

457. You receive a referral from an osteopath. In this situation:

a) you need a referral from a non-osteopathic primary care physician.
b) **you can proceed with the patient examination**.
c) both the osteopathic and a primary care physician referral are required to begin the examination.
d) you cannot accept referrals originating from osteopaths and must refuse to examine the patient at this time.

458. A patient with a past medical history of recurrent pneumonia is referred to physical therapy with new symptoms of cough and sputum production. The examination technique that would provide the least important information in this case is:

a) Sa02 measurement.
b) auscultation.
c) **thoracic excursion**.
d) vital sign measurement.

459. During an adaptive seating clinic, you are asked to examine a child who is wheelchair-bound. Your findings include asymmetrical shoulders and a forward, flexed head. Based on these findings, the next thing you would do is:

a) recommend an abduction wedge for the chair.
b) **check to see if the pelvis is symmetrical and supported**.
c) recommend a high back with head supports for the chair.
d) a scoliosis screening

460. A therapist that you supervise has been arriving 10-15 minutes late each day for the past week. The clinic hours are 8:00AM to 5:00 PM. This tardiness occurs on Monday of the following week. In this case you should:

a) discuss the issue with the therapist during lunch in the cafeteria.
b) say nothing as the therapist usually stays until 5:20 each day.
c) ask the therapist to meet with you immediately.
d) **ask the therapist to meet with you during a schedule break.**

461. The balance test which specifically discriminates for vestibular dysfunction is the:

a) **Clinical Test for Sensory Interaction in Balance**.
b) Berg Balance Test.
c) Timed Up and Go Test.
d) Multidirectional Reach Test.

462. Your preprosthetic examination of a patient with a transfemoral amputation reveals arching of the lumbar spine when in the supine position. It is most likely this is a result of:

a) tight hip extensors and weak abdominals.
b) tight back extensors and weak hip extensors.
c) weak back extensors and weak hip flexors.
d**) tight hip flexors and weak abdominals**.

463. When designing the hydrotherapy area in a private practice, the electrical components critical to include are:

a) rheostats.
b) three-slotted electrical outlets.
c) surge protectors.
d) **wall outlets with ground fault interrupters**.

464. You are examining an eight year-old girl with spastic diplegia. An indication that the asymmetrical tonic neck reflex is integrated would be if the child can:

a) turn her head from side to side as the shoulders alternately extend.
b) maintain gaze at both hands held in front.
c) ventroflex the neck while both shoulders flex.
d) **turn her head from side to side with no corresponding limb movement.**

465. To improve abduction and external rotation of the shoulder, the best diagonal PNF pattern to use is:

a) D1 flexion.
b) **D2 flexion**.
c) D1 extension.
d) D2 extension.

466. A patient with peripheral vascular disease has an ulcer located on the medial malleolus. This is most likely indicative of:

a) an indurated ulcer.
b) a pressure ulcer.
c) **venous insufficiency**.
d) arterial insufficiency.

467. A women with a fixed deformity of minus ten degrees of plantar flexion would have the greatest difficulty:

a) walking with one inch heels.
b) walking barefoot.
c) **ascending a ramp**.
d) walking down stairs.

468. To improve artistic performance, you should advise an opera singer to strengthen the:

a) oblique muscles.
b) abdominal muscles.
c) **muscles of inspiration**.
d) muscles of expiration.

469. Following trauma, extreme care must be taken in the management of myositis ossificans. The muscle most often affected is the:

a) **brachialis.**
b) biceps brachii.
c) quadriceps femoris.
d) triceps.

470. A classical sign indicative of congenital hip dysplasia in the newborn is:

a) **limited passive abduction.**
b) limited external rotation.
c) excessive internal rotation.
d) limited passive adduction.

471. A child with brain damage shows evidence of mental retardation, tactile defensiveness and proximal joint instability. Activities that the physical therapist can best utilize in this case include:

a) neutral warmth and rhythmic stabilization.
b) repetitive brushing and joint traction.
c) very light touch and weightbearing.
d) **firm pressure and joint approximation**.

472. During a phonophoresis treatment of the shoulder, a patient reports to you that he is experiencing a deep aching pain. This situation is best remedied by:

a) **adding more coupling agent and moving the sound head more rapidly**.
b) reducing the power and moving the sound head over a large area.
c) using a smaller sound head but doubling the exposure time.
d) decreasing the frequency and moving the sound head more rapidly.

473. In ordering a wheelchair for someone with pronounced flexor spasticity, the components which ordinarily should not be included are:

a) **elevating leg rests.**
b) removable arm rests.
c) detachable, swing-away foot rests.
d) web heel loops.

474.
475. In order to differentiate between pain originating in the sacroiliac joints as opposed to hip pathology, it would be best to employ the:

a) **Fabere Test.**
b) compression-distraction tests.
c) Noble Compression Test.
d) straight leg test.

476. You are to treat a patient with a large posterolateral protrusion of a herniated nucleus pulposus. Examination reveals decreased sensation, radiating pain and 1+ Achilles DTR in the left lower extremity. Trunk flexion is limited to 45 degrees and pain increases with sitting of more than 15 minutes. Based on this symptomatology, it would be best if treatment stressed:

a) prone and standing trunk extensions followed by hip shifting to the left.
b) Williams flexion exercises, posterior pelvic tilting and hamstring stretch.
c) posterior pelvic tilting, bridging, diagonal curl ups and gradual increase in

477. sitting tolerance.
d) **hip shifting to the right followed by prone and standing trunk extensions.**

476. A patient with chronic rheumatoid arthritis lacks self-esteem and seems uninterested during physical therapy treatment session. In this case:

a) **get the patient involved in making choices regarding day to day treatment options**.
b) remind the patient that others with similar disorders have greater limitations.
c) keep the patient fully informed of the therapeutic outcomes you envision.
d) use appropriate behavior modifications if there are any signs of depression or withdrawal.

477. You have evaluated a nine month-old who cannot assume or maintain the quadruped position without assistance. The parents insist that the child has already begun to walk with assistance. You suspect that what the parents are describing is most likely the result of:

a) protective extension downward.
b) **spontaneous stepping**.
c) the positive supporting reaction.
d) the negative supporting reaction.

478. You are treating a 16 year-old female athlete for patellofemoral malalignment syndrome. During competition, the orthosis that would be most helpful in aiding this situation is a:

a) knee orthosis with a hinge joint.
b) derotation brace.
c**) patellar stabilizing brace**.
d) knee orthosis with an offset hinge joint.

479. A construction worker seeks physical therapy treatment for pain and weakness in the right upper extremity. The physical therapy examination of the right upper extremity reveals the pain distribution to be in the lateral forearm, weakness in elbow flexion and wrist extension and a decreased brachioradialis reflex. Based on these findings, the next thing the physical therapist should do is:

a) **a cervical screening examination**.
b) institute a program of muscle strengthening and sensory facilitation for the involved areas.
c) check upper torso and neck postural alignment.
d) immediate referral to a physician for EMG studies.

480. Exercise guidelines for the well-trained, well-controlled insulin dependent athlete with diabetes recommend all of the following except:

a) decreasing or omitting the insulin dose just following the exercise or athletic activity.
b) decreasing or omitting the insulin dose just prior to exercise or athletic activity.
c) **decreasing or omitting carbohydrate snacks just following intense exercise or activity**.
d) ingesting up to 30 grams of additional carbohydrate for every 30 minutes interval of intense exercise or activity.

481. A resting pulse rate of 40-60 beats per minute would unlikely to be found in:

a) marathon runners.
b) patients taking digitalis.
c) elderly individuals with arteriosclerotic hearth disease.
d) **children under five years of age.**

482. A patient developed tarsal tunnel syndrome secondary to a pre-existing pes valgus. All of the following shoe modifications might help to correct this problem except for a:

a) heel insert.
b**) rocker sole**.
c) scaphoid pad.
d) medial heel wedge.

483. Six weeks following a Smith's fracture the cast is removed when healing is complete. A rehabilitation program, including joint mobilization, is to be instituted. In this case, the mobilization techniques that would be of most benefit to enhance joint play include:

a) anterior and posterior glides of the radioulnar joint.
b) **anterior and posterior glides of the radiocarpal joint**.
c) medial and lateral glides of the radiocarpal joint.
d) joint traction at the midcarpal joint.

484. The only activity which is appropriate for a physical therapist assistant to perform is to:

a) alter the treatment plan if, in the assistant's judgment, it would benefit the patient.
b) discharge a patient from home physical therapy care if the assistant feels that the patient has achieved maximum benefit.
c) write the discharge plan for the patient.
d) **immediately discontinue any treatment procedures which, in the assistant's judgment, appears to be harmful to the patient.**

485. You are examining the shoulder of a referred patient. The patient experiences maximal pain after 150 degrees of active abduction. This pain pattern is most characteristic of:

a) glenohumeral subluxation.
b) subacromial bursitis.
c) supraspinatus tendinitis.
d) **acromioclavicular sprain**.

486. One of your evaluative findings of a patient reveals hypoesthesia in the midpalmer region and the terminal phalanges of the index and middle fingers. These symptoms are most characteristic of:

a) Guillain-Barré Syndrome.
b) scalenus anticus syndrome.
c) **carpal tunnel syndrome**.
d) alcoholic peripheral polyneuropathy.

487. A 17 year-old gymnast has been diagnosed with an unstable spondylolisthesis involving L5 on S1. Appropriate management of this problem by a physical therapist should include:

a) back extension exercises.
b) **abdominal strengthening and flexion exercises.**
c) joint mobilization emphasizing lumbar rotation in both supine and sidelying.
d) lumbar extension mobilization using hand pressure directed anteriorly.

488. Following a thoracotomy, a patient developed right-sided atelectasis.

To improve oxygenation to the right lung, it would be best to position this patient:

a) **left sidelying**.
b) supine.
c) prone.
d) right sidelying.

489. A 6 year-old child with myelomeningocele at the L5 level and minimal complications should be expected to ambulate with:

a) a reciprocating gait orthosis with a Rollator walker.
b) a shoe insert to maintain proper foot alignment and no ambulatory aid.
c) a knee-ankle-foot orthosis with crutches.
d) **an ankle-foot orthosis with crutches or a Rollator walker**.

490. If a patient has longstanding rheumatoid arthritis of the cervical spine, the complication that is most likely to be seen and of concern is:

a) facet joint impingement.
b) disk herniation.
c) spinal stenosis.
d) **subluxation.**

491. A patient enters your private practice with a chief complaint of low back pain. He had been treated with chemotherapy for adrenal carcinoma about two years ago and claims to be disease free. Your examination also reveals a history of recent episodes of urinary incontinence. In this case, you should:
a) treat the patient appropriately for his back pain which may also help to correct the incontinence.
b) **not begin treatment and have the patient see his primary care physician immediately**.
c) treat the patient for back pain and refer him to his primary care physician if symptoms do not improve after one month.
d) refuse to treat the patient since you believe the recent back pain might be a result of metastatic disease.

492. A patient with hemiplegia, who uses an ankle-foot orthosis, is experiencing recurvatum and toe drag during ambulation To correct these problems, adjustments should be made to the:

a) **posterior stop.**
b) knee locking mechanism.
c) anterior stop.
d) metal calf band.

493. A patient with a C6 lesion is referred for physical therapy after being fit with a halo device. The first functional mat activity to work on would be:

a) sliding board transfer with assistance.
b) **rolling.**
c) stand-pivot-sit transfer with assistance.
d) sitting balance.

494. You are examining a patient with facet impingement in the right lumbar region who complains of pain which follows a dermatomal distribution. When upright, the patient assumes a protective posture of left sidebending and right trunk rotation. Pain and restriction of motion increase when you ask the patient to move in the opposite direction. This opposite motion results in:

a) both facets opening.
b) both facets closing.
c) **closing of the right facet and opening of the left.**
d) opening of the right facet and closing of the left.

495. A patient with a spinal cord injury has accomplished all anticipated goals in the plan of care. The patient is to be discharged from the rehabilitation center; however, the family insists that the patient is not ready to go home. They are willing to pay out-of-pocket for additional physical therapy. As the physical therapist working with this patient you should:

a) reformulate the discharge plan to reflect a change in the plan of care.
b) continue with maintenance activities if the family pays.
c) have the family discuss additional physical therapy with the physician-in-charge.
d) **discharge the patient from physical therapy.**

496. An obese, 57 year-old man with a long history of heavy smoking and alcoholic abuse was admitted with a diagnosis of bronchitis. Some findings included copious green sputum production, dyspnea, coarse rales in both lower lobes and severe deconditioning. PaO2 was 62mmHg and FEV1:FVC ratio was 39%. At discharge, though medically stable, his basic symptoms persisted. The physical therapy discharge plan should have as its highest priority short term goal:

a) improvement in breathing patterns including mechanics, control and relaxation.
b) education to decrease smoking, alcoholic intake and weight.
c) **an increase in airway patency via self-administered bronchopulmonary hygiene**.
d) aerobic exercise of , at least, 20 consecutive minutes per day to help decrease shortness of breath

497. During phase II of a cardiac rehabilitation program, an appropriate activity to meet the metabolic goals of this treatment level would be:

a) walking on a level surface at 2.5 miles per hour.
b) **walking on a level surface at 4.5 miles per hour.**
c) stationary bicycle riding at 5 miles per hour.
d) exercise with 2 or 3 pound weights.

498. Your patient sustained an intertrochanteric fracture of the femur as a result of a fall. She underwent open reduction, internal fixation three weeks ago. In an effort to better manage her pain, you began a program of very slow ambulation in a therapeutic pool with the water level chest high. You now have permission to increase weightbearing to tolerance. To initiate this change of treatment in the pool you would ambulate the patient:

a) at the same water depth, but increase the velocity of ambulation.
b) in deeper water.
c) **in shallower water.**
d) at the same water depth, but have the patient wear ankle weights.

499. You are requested to take a culture from a partial thickness burn wound. The proper technique to follow when collecting the specimen is to swab the:

a) wound exudate in the area of the immediate surrounding tissue.
b) wound and immediate surrounding tissue.
c) wound, immediate surrounding tissue and any exudate on nearby bed linen.
d) **actual wound site only**.

500. You are using electrical stimulation as a tool to help facilitate and re-educate various muscle groups The patient is hypersensitive to electrical stimulation. The current parameters that would be best to use in this case are:

a) interrupted monophasic with a wide pulse width and low intensity.
b) high voltage, monophasic twin-spiked.
c) **interrupted biphasic with a wide pulse width and low intensity.**
d) Russian stimulation modulated to 70 pulses per second.